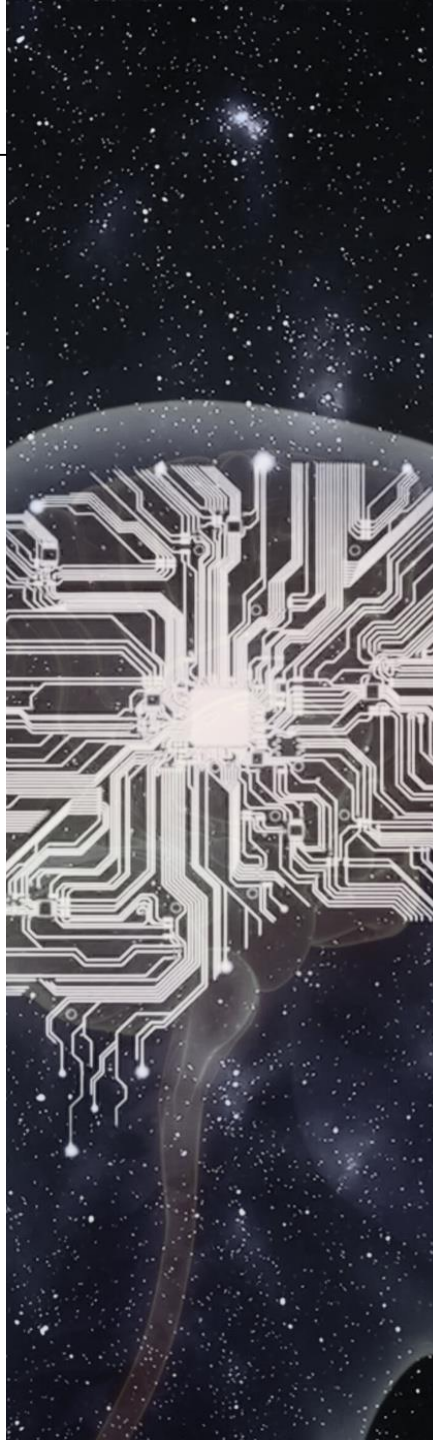
어셈블리프로그래밍

2. x86 Processor architecture



General concepts

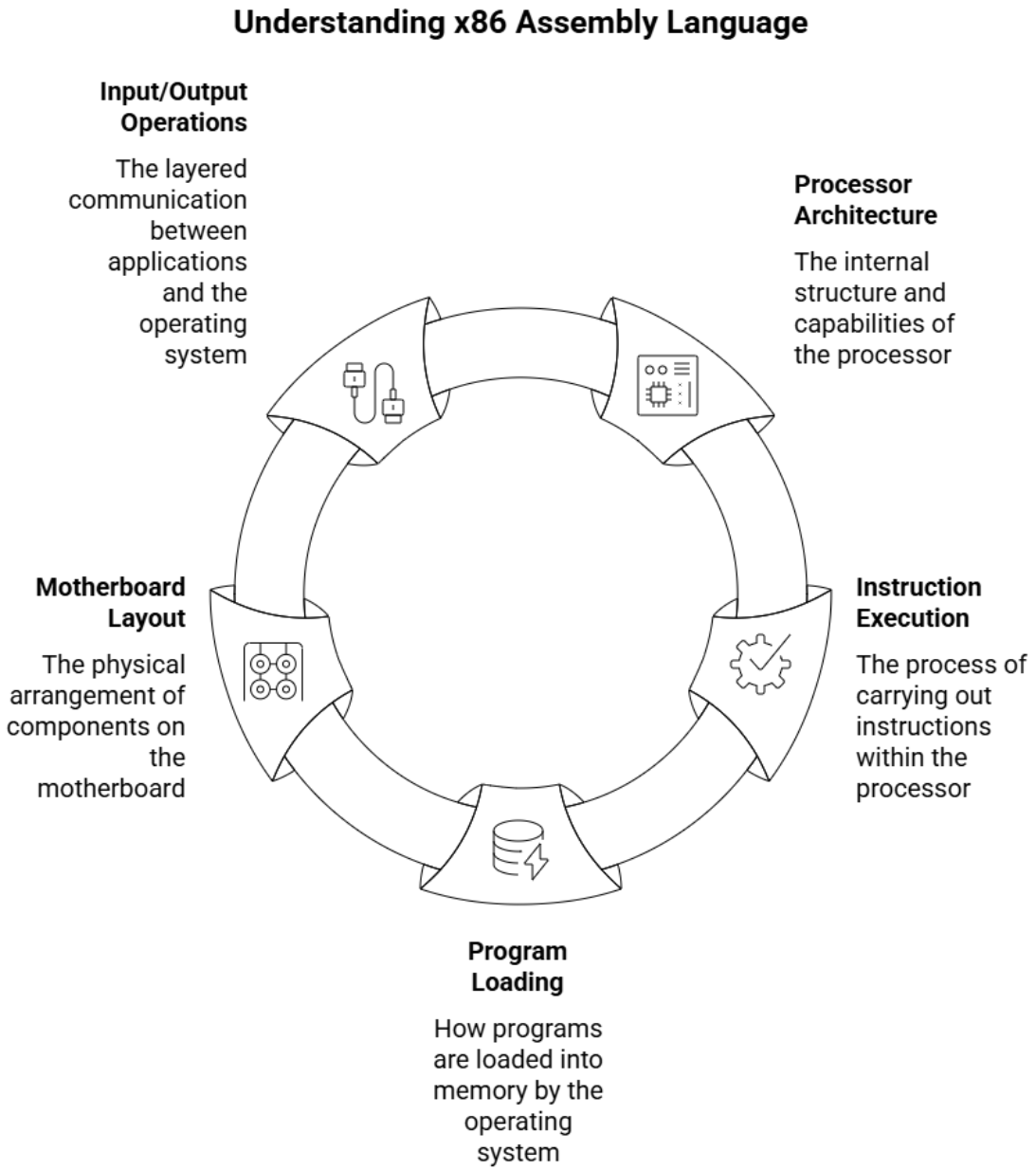
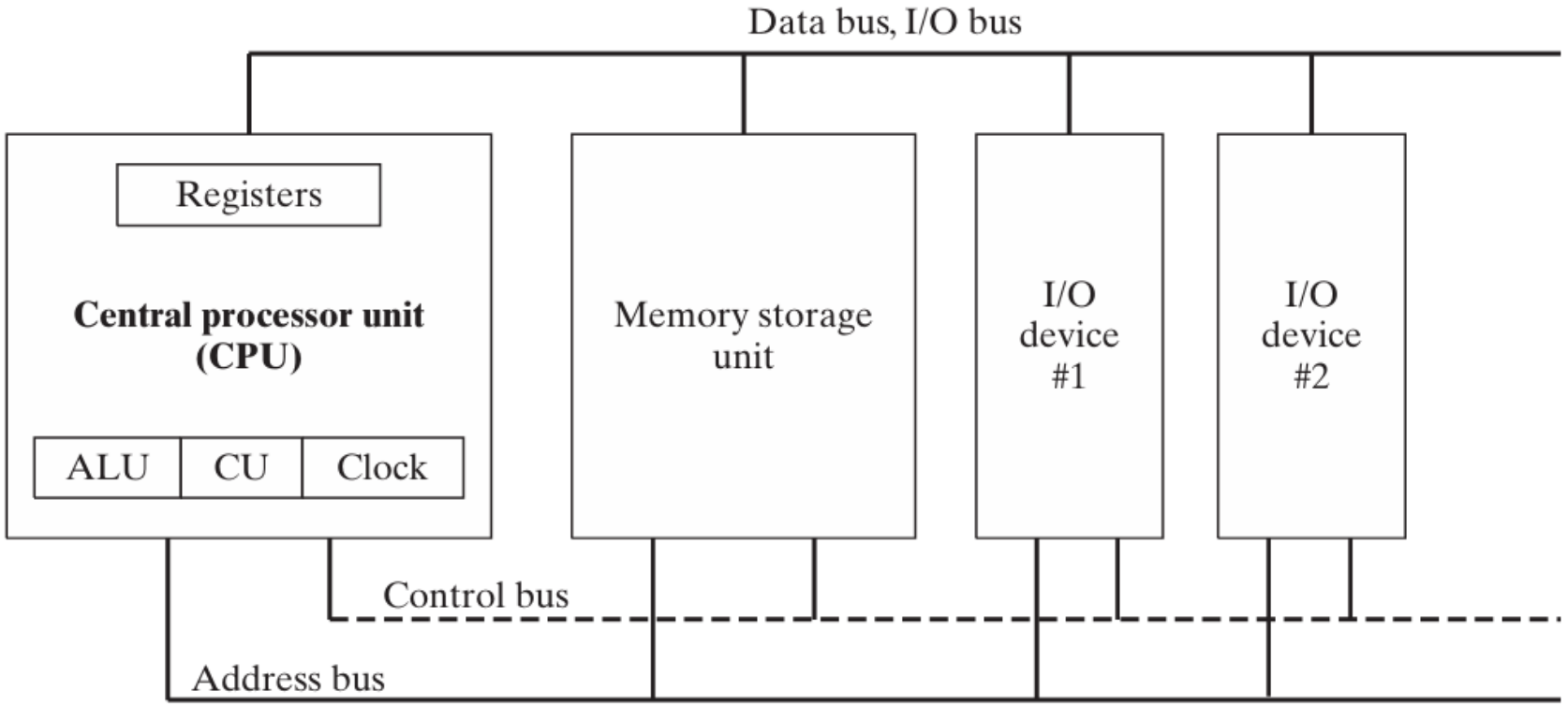
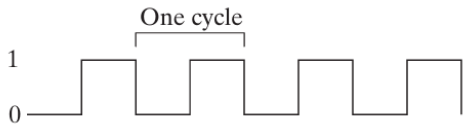
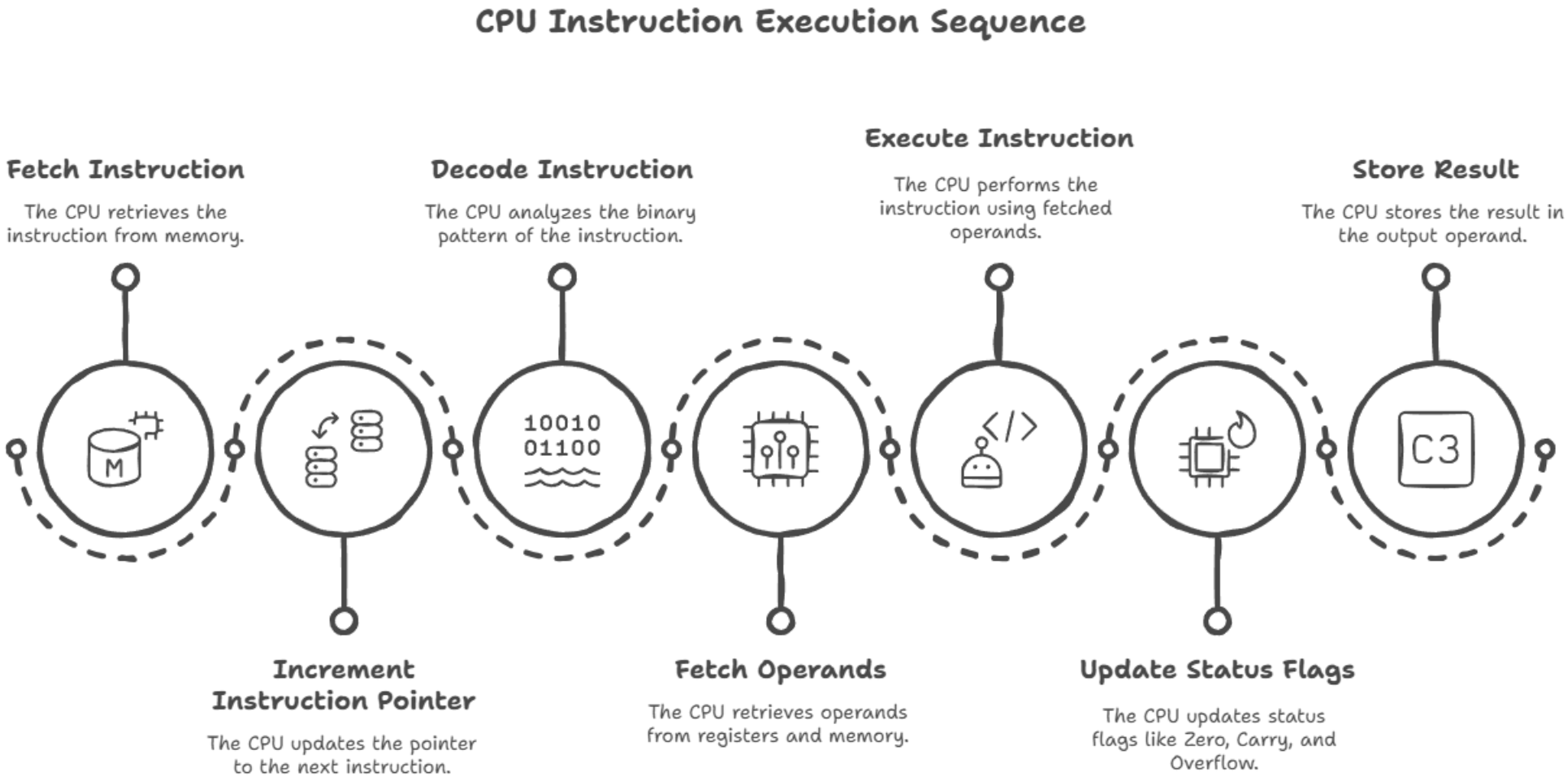


FIGURE 2–1 Block diagram of a microcomputer.






CPU Instruction Execution Sequence
Fetch Instruction
The CPU retrieves the instruction from memory.
M
Increment Instruction Pointer
The CPU updates the pointer to the next instruction.
Decode Instruction
The CPU analyzes the binary pattern of the instruction.
10010
01100
Fetch Operands
The CPU retrieves operands from registers and memory.
Execute Instruction
The CPU performs the instruction using fetched operands.
Update Status Flags
The CPU updates status flags like Zero, Carry, and Overflow.
Store Result
The CPU stores the result in the output operand.
C3
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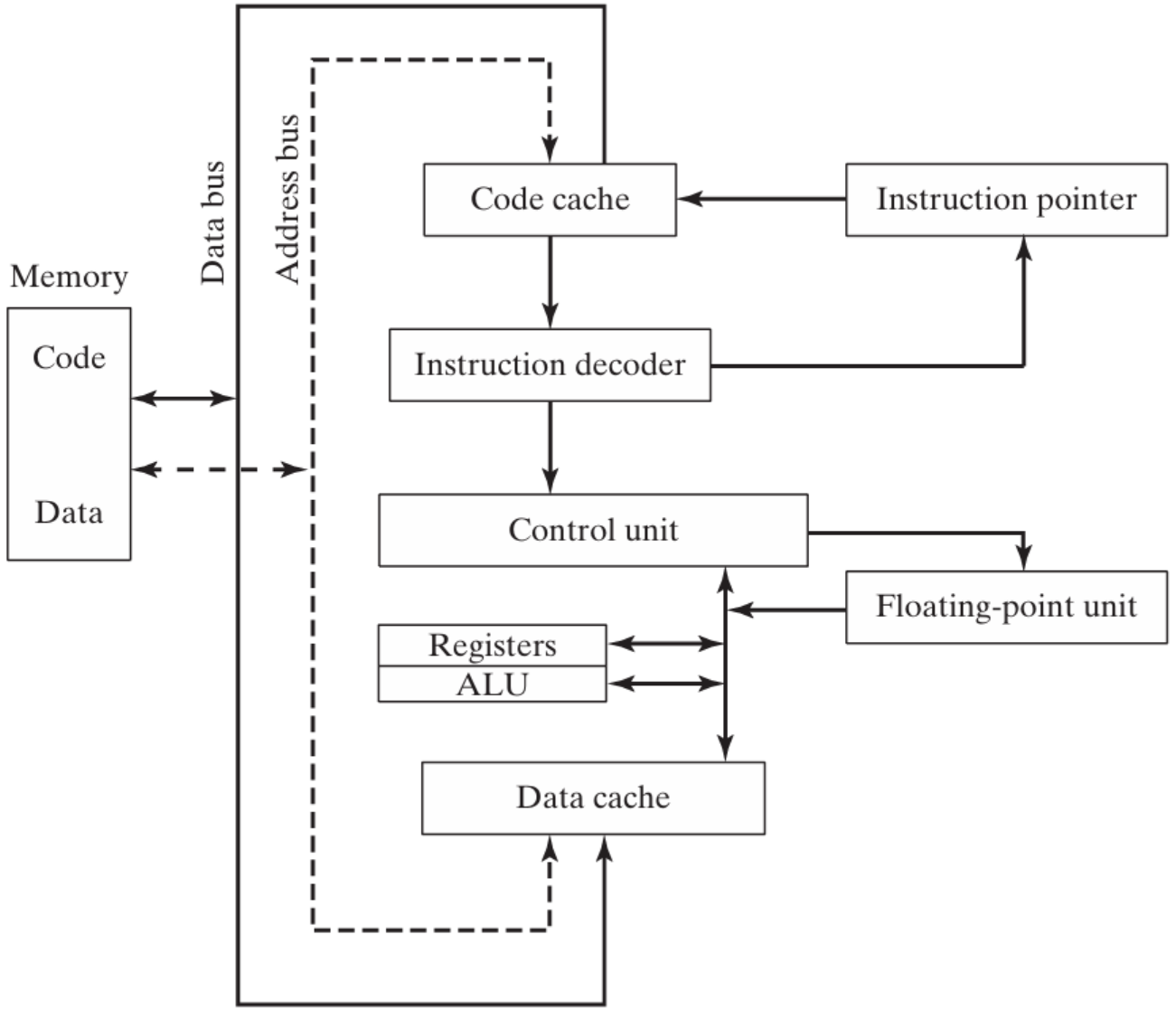
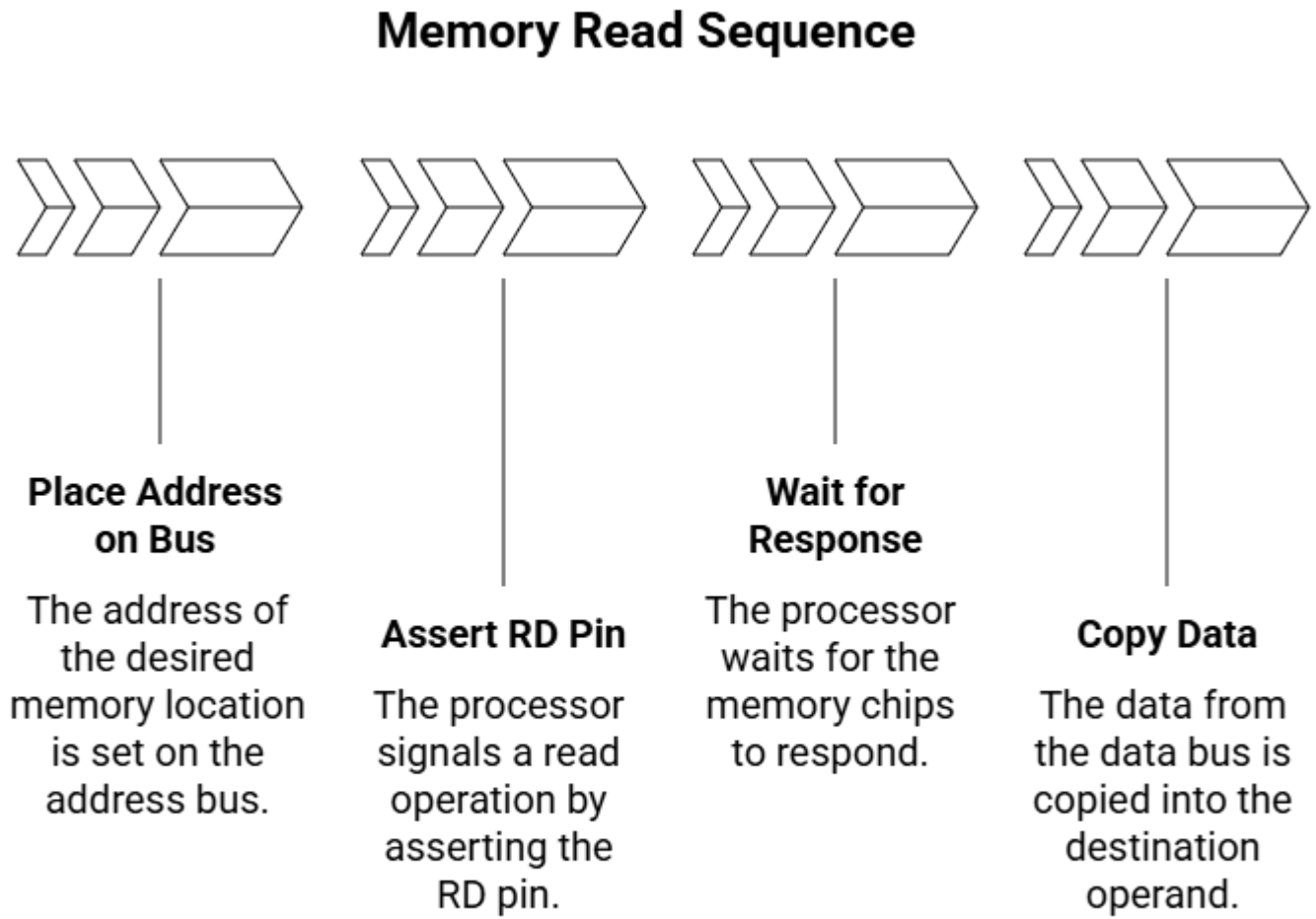
🕑 Instruction Decode Timeline (예: `MOV AX,1234h` = `B8 34 12`)

1. **Fetch B8**
   - IP = 1000 → 1001
   - 첫 바이트(`B8`) 가져오기
2. **Decode opcode (B8)**
   - CPU는 `MOV AX, imm16` 명령임을 인식
   - → **즉시 2바이트 operand 필요**하다는 걸 알게 됨
3. **Fetch 34**
   - IP = 1001 → 1002
   - 첫 번째 operand 바이트 가져오기
4. **Fetch 12**
   - IP = 1002 → 1003
   - 두 번째 operand 바이트 가져오기
5. **Instruction complete**
   - 이제 `B8 34 12` 전체 명령어 해독 완료
   - `MOV AX,1234h`라고 확정
6. **Execute**
   - AX ← 1234h

Figure 2–2 Simplified CPU block diagram.




Memory Read Sequence
Place Address on Bus
The address of the desired memory location is set on the address bus.
Assert RD Pin
The processor signals a read operation by asserting the RD pin.
Wait for Response
The processor waits for the memory chips to respond.
Copy Data
The data from the data bus is copied into the destination operand.
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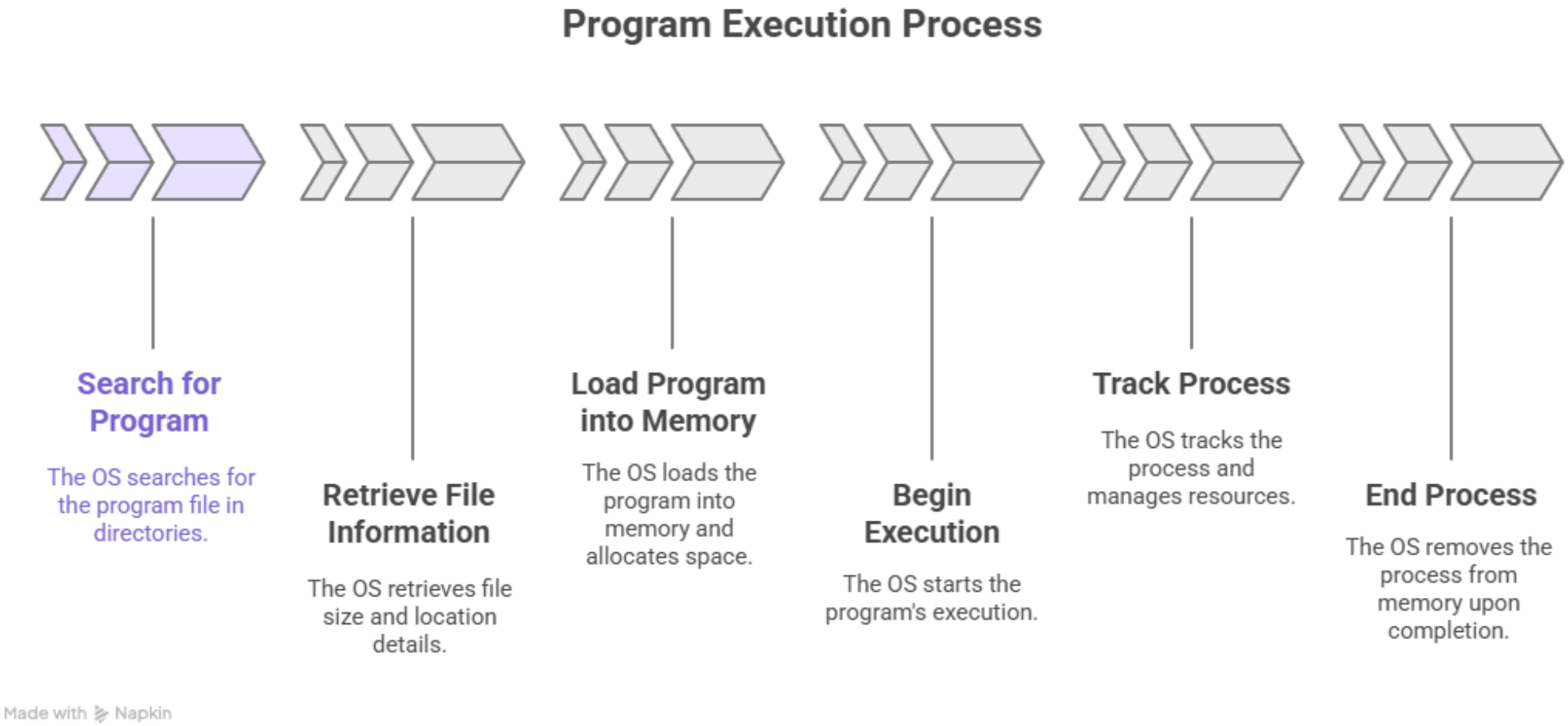
Program Execution Process
Search for Program
The OS searches for the program file in directories.
Retrieve File Information
The OS retrieves file size and location details.
Load Program into Memory
The OS loads the program into memory and allocates space.
Begin Execution
The OS starts the program's execution.
Track Process
The OS tracks the process and manages resources.
End Process
The OS removes the process from memory upon completion.
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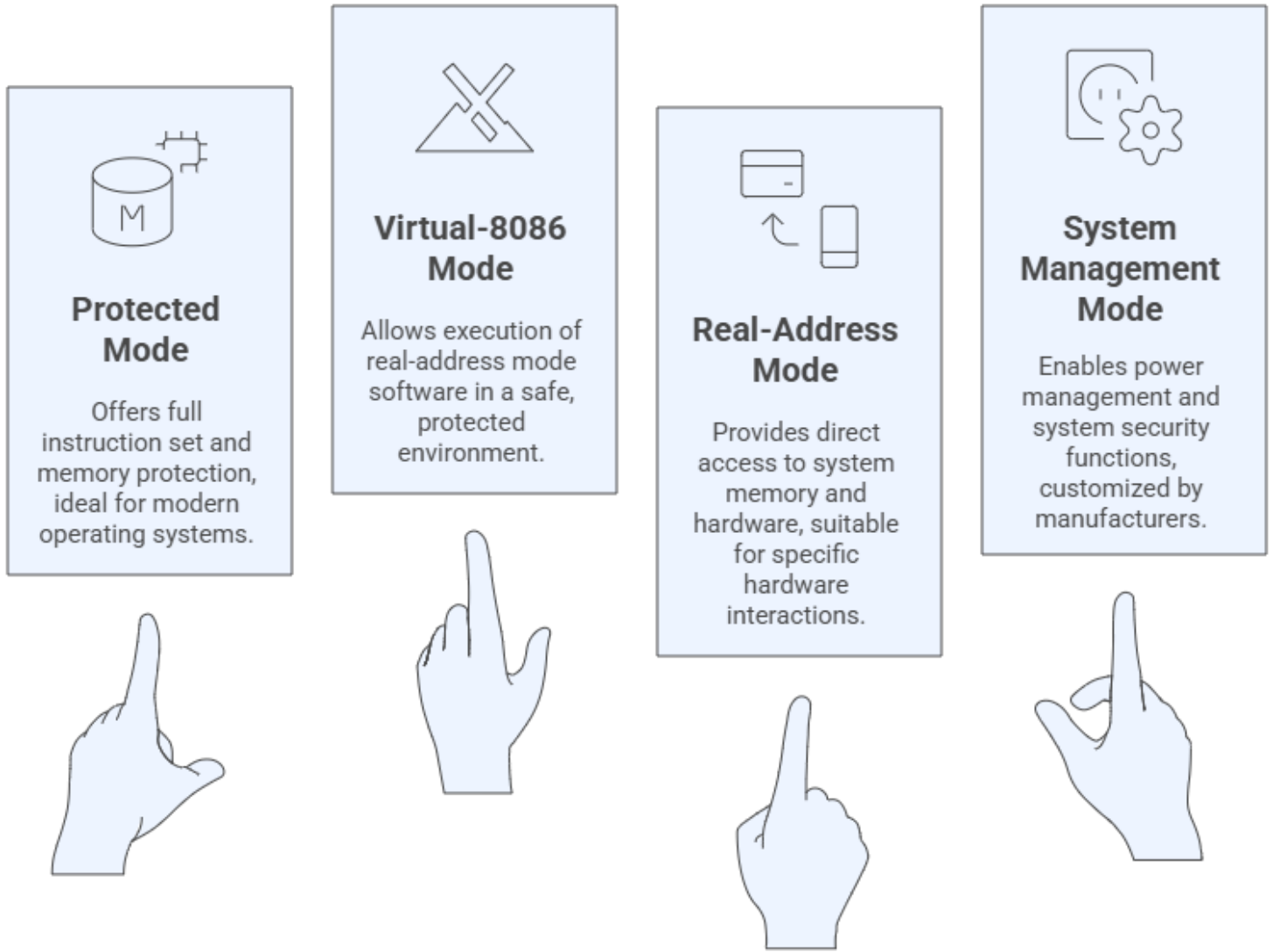
Which x86 processor mode should be used?



Memory Addressing Capabilities

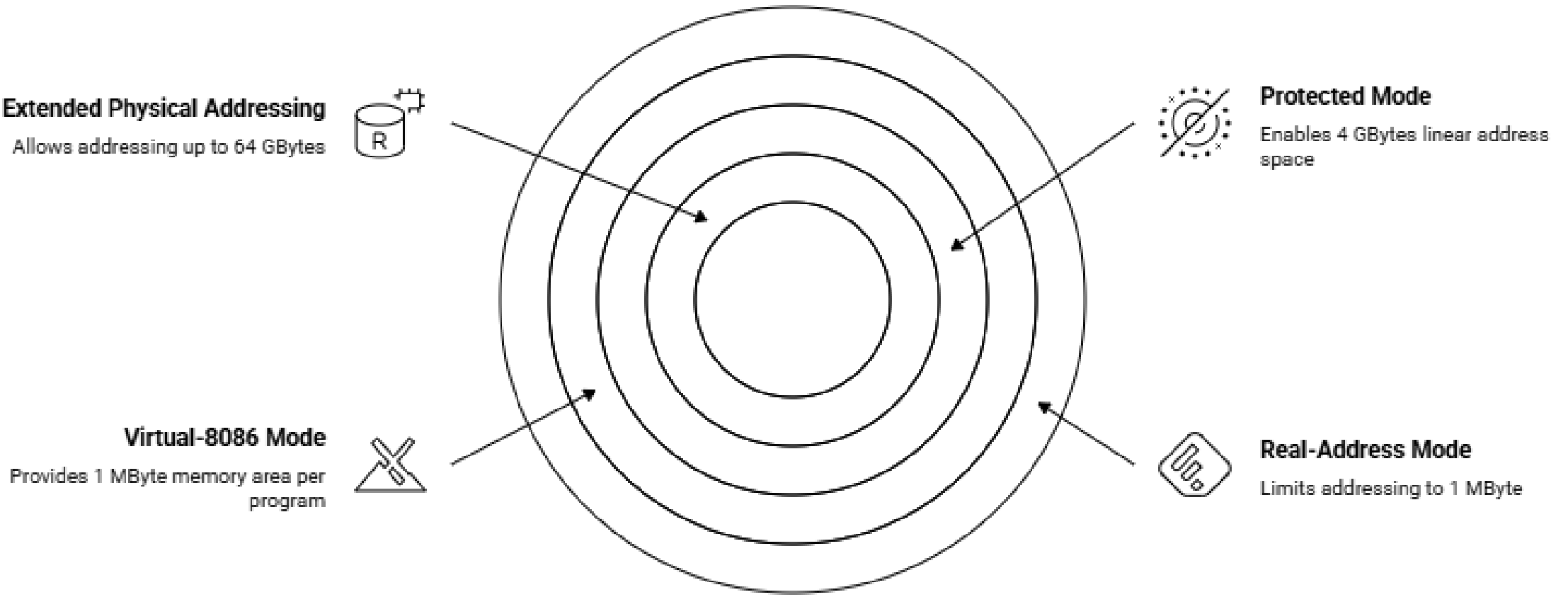


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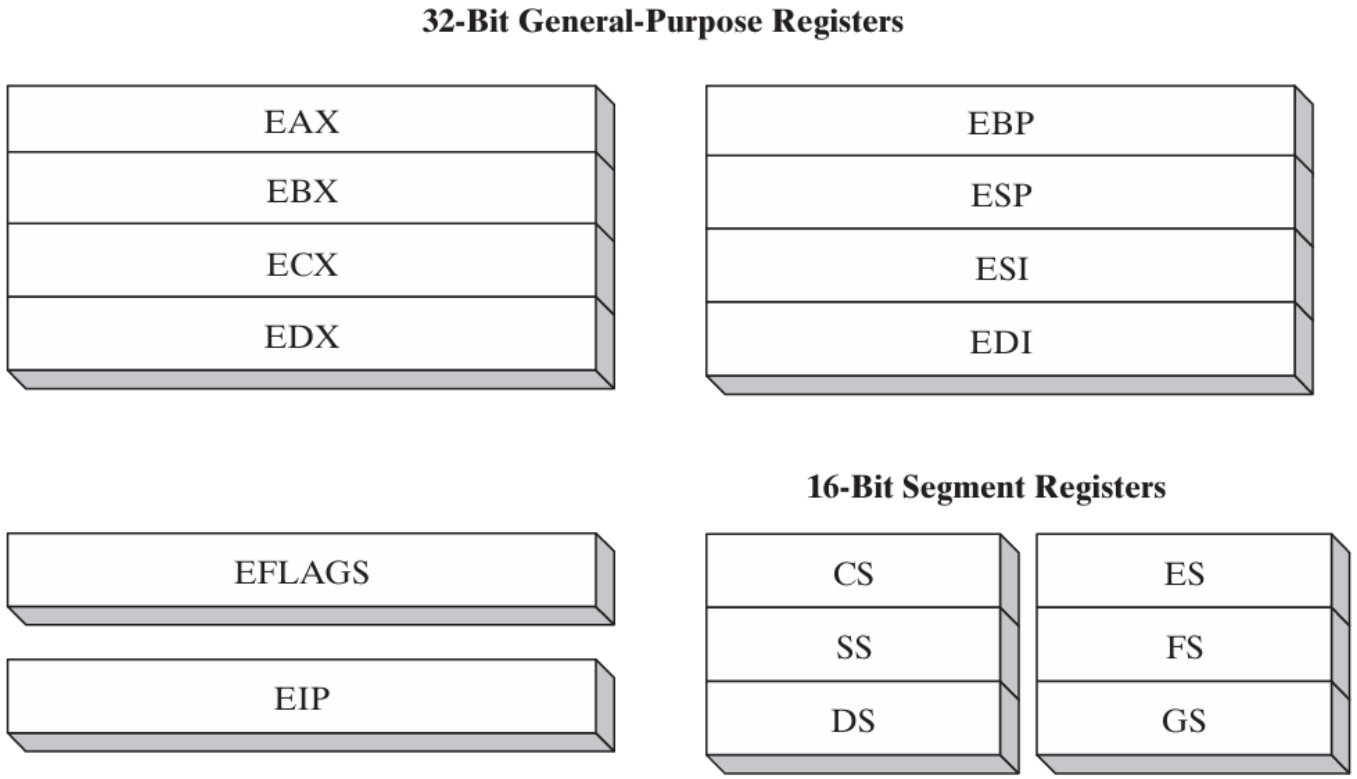
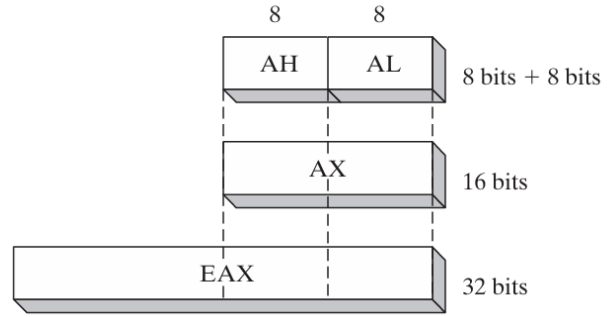
Figure 2–3 Basic program execution registers.



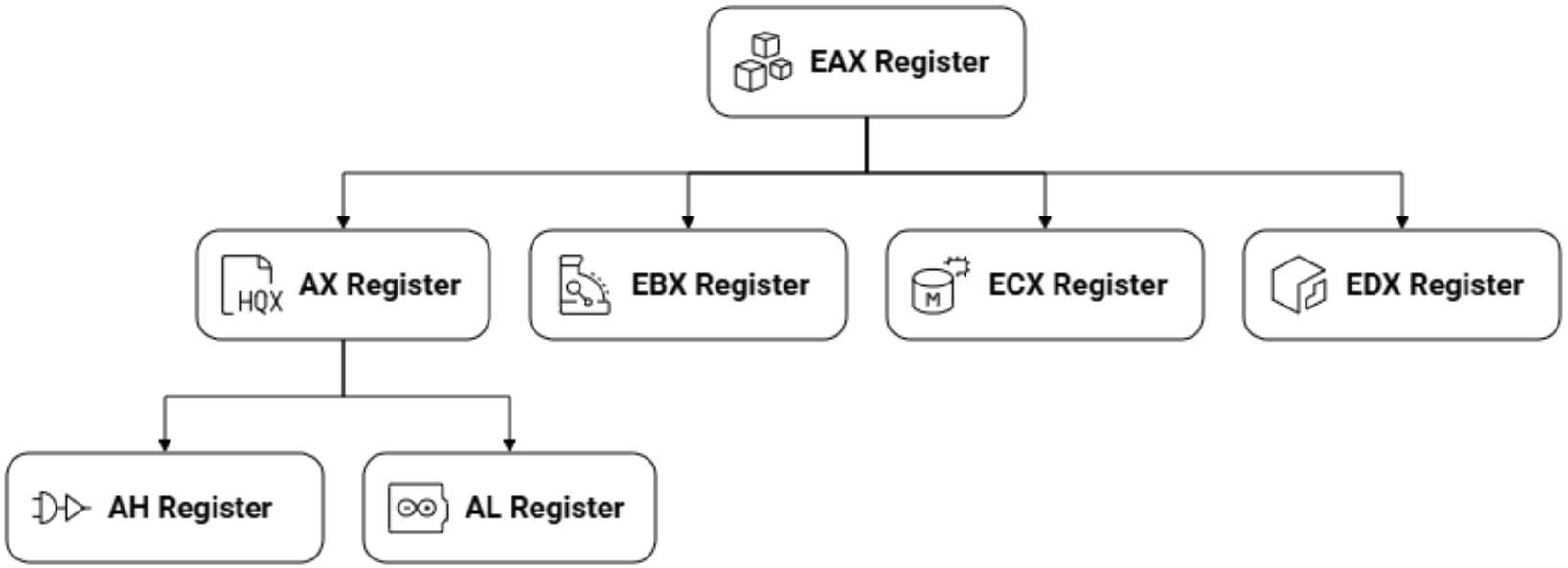
32-Bit	16-Bit	8-Bit (High)	8-Bit (Low)
EAX	AX	AH	AL
EBX	BX	BH	BL
ECX	CX	CH	CL
EDX	DX	DH	DL

32-Bit	16-Bit
ESI	SI
EDI	DI
EBP	BP
ESP	SP

Figure 2–4 General-purpose registers.



General-Purpose Registers and Their Subdivisions



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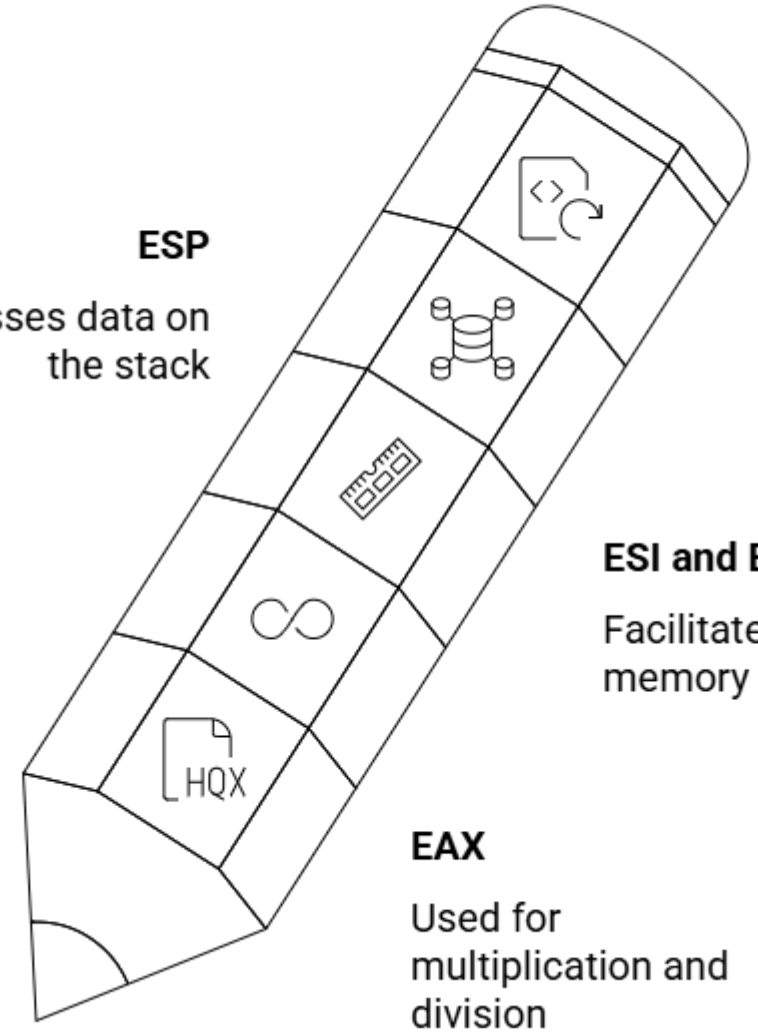
Specialized Register Functions



ESP

Addresses data on the stack

EBP

References function parameters and local variables

ECX

Serves as a loop counter

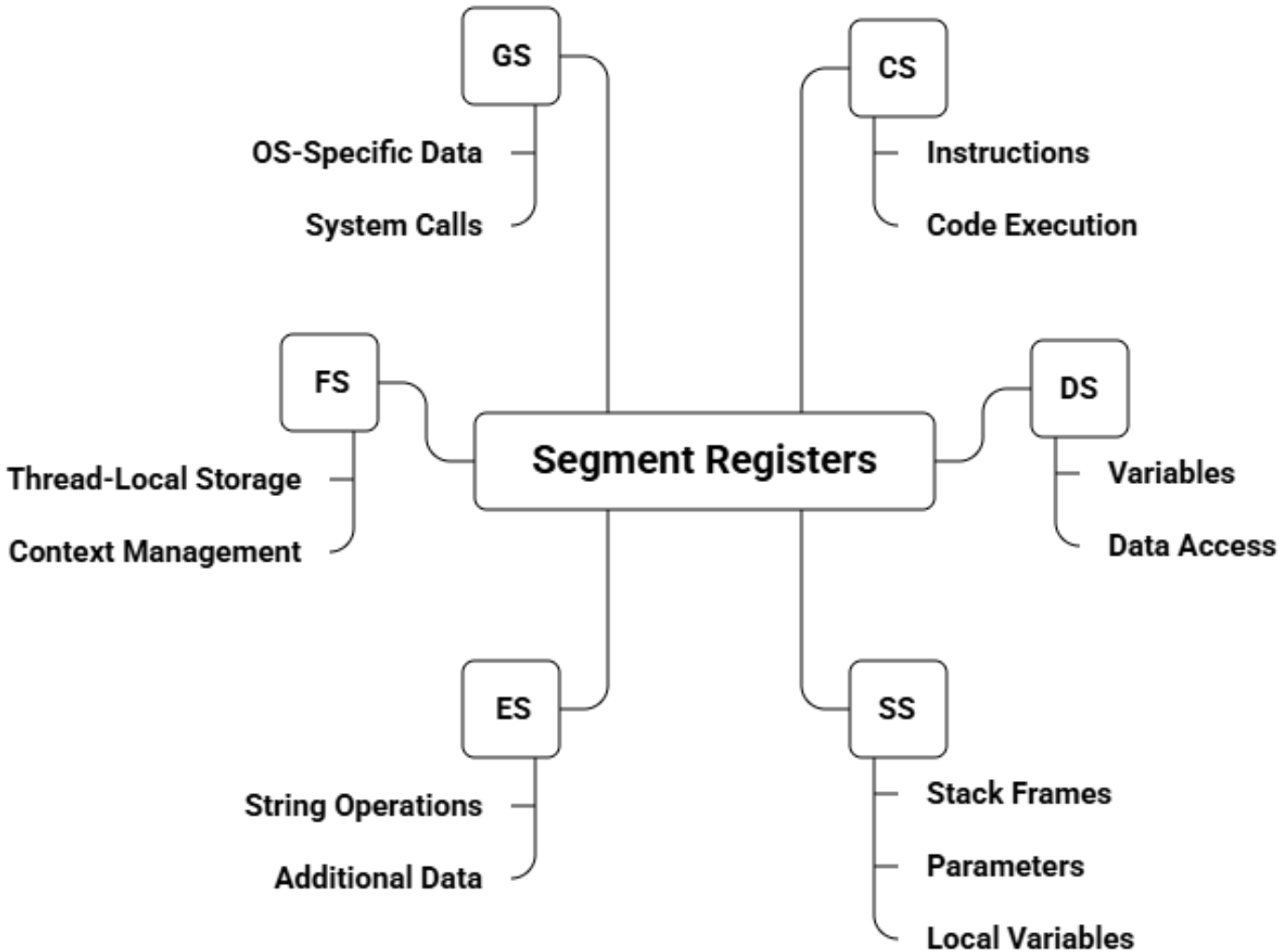
ESI and EDI

Facilitate high-speed memory transfers

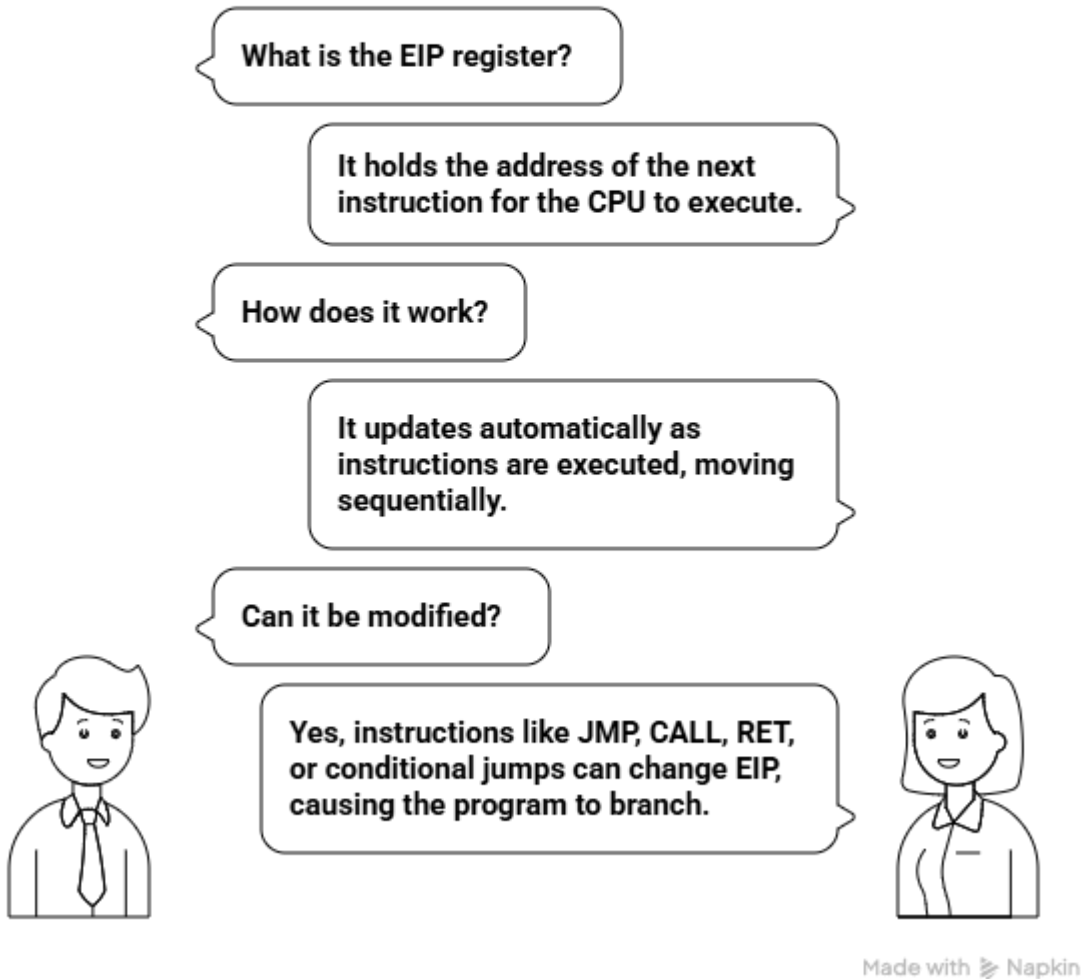
EAX

Used for multiplication and division

x86 Segment Registers and Their Functions



EIP Register Explained



Comparison of CPU Status Flags

Status Flag	Description
Carry Flag (CF)	Unsigned result too large for destination
Overflow Flag (OF)	Signed result too large/small for destination
Sign Flag (SF)	Result is negative
Zero Flag (ZF)	Result is zero
Auxiliary Carry Flag (AC)	Carry from bit 3 to bit 4 (8-bit)
Parity Flag (PF)	Even number of 1 bits in result

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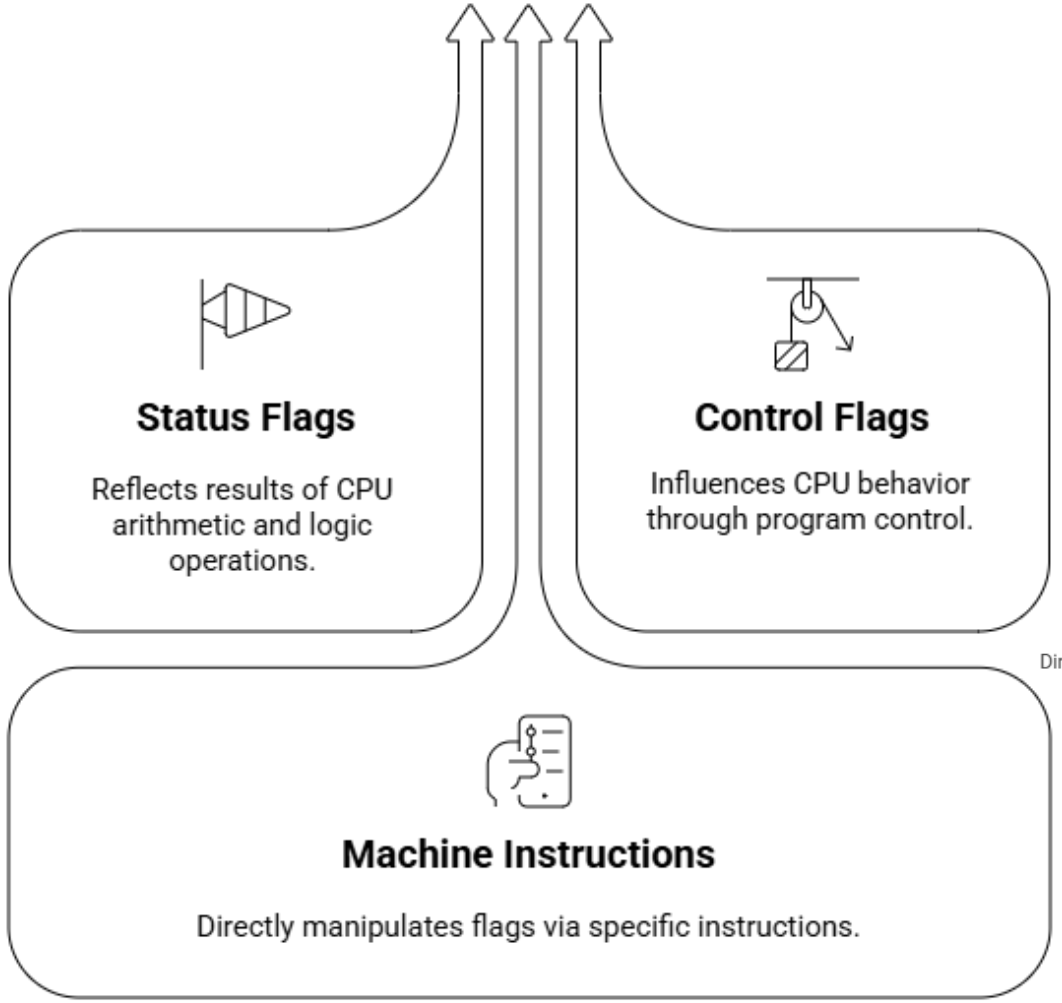
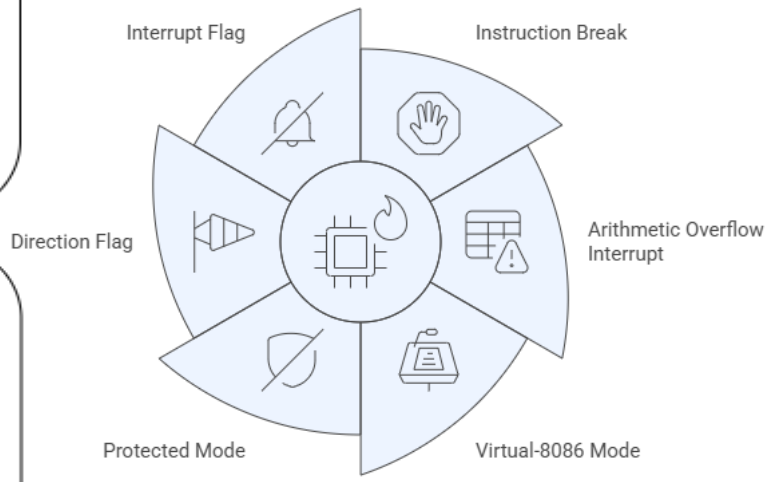
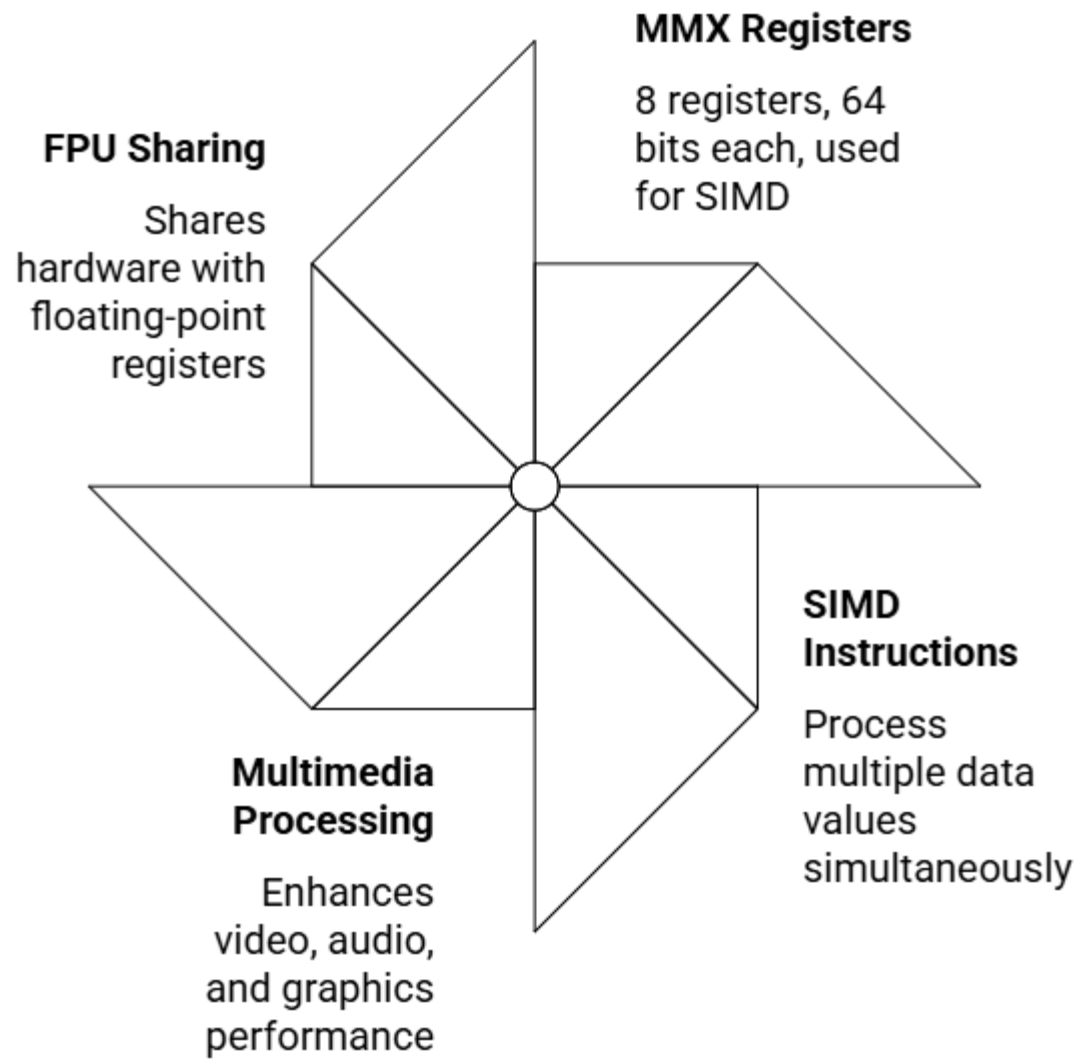
EFLAGS Register
Flags and CPU Control



CPU Control Flags Overview

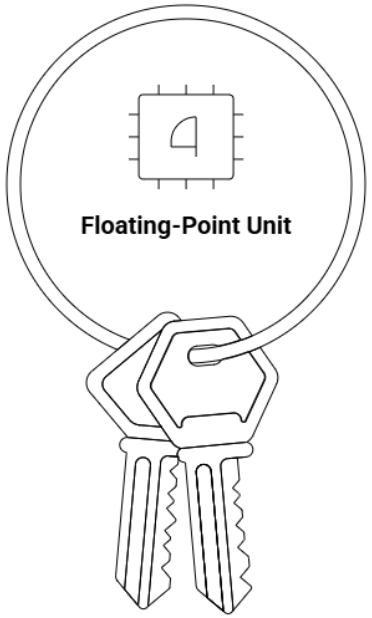


MMX Technology Overview



32-Bit x86 Processors

FPU's Role in Arithmetic



High-Speed Execution

The FPU's ability to perform calculations quickly.

Floating-Point Arithmetic

The specific type of calculations the FPU handles.

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Evolution of Floating-Point Unit Integration

Separate Coprocessor Chip

The FPU was initially a separate chip.

Intel486 Integration

The FPU was integrated into the main processor chip.

Modern Processor Design

Modern processors include the FPU as a standard component.

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FPU Data Registers

ST(0)
First floating-point data register.

ST(1)
Second floating-point data register.

ST(2)
Third floating-point data register.

ST(3)
Fourth floating-point data register.

ST(4)
Fifth floating-point data register.

ST(5)
Sixth floating-point data register.

ST(6)
Seventh floating-point data register.

ST(7)
Eighth floating-point data register.

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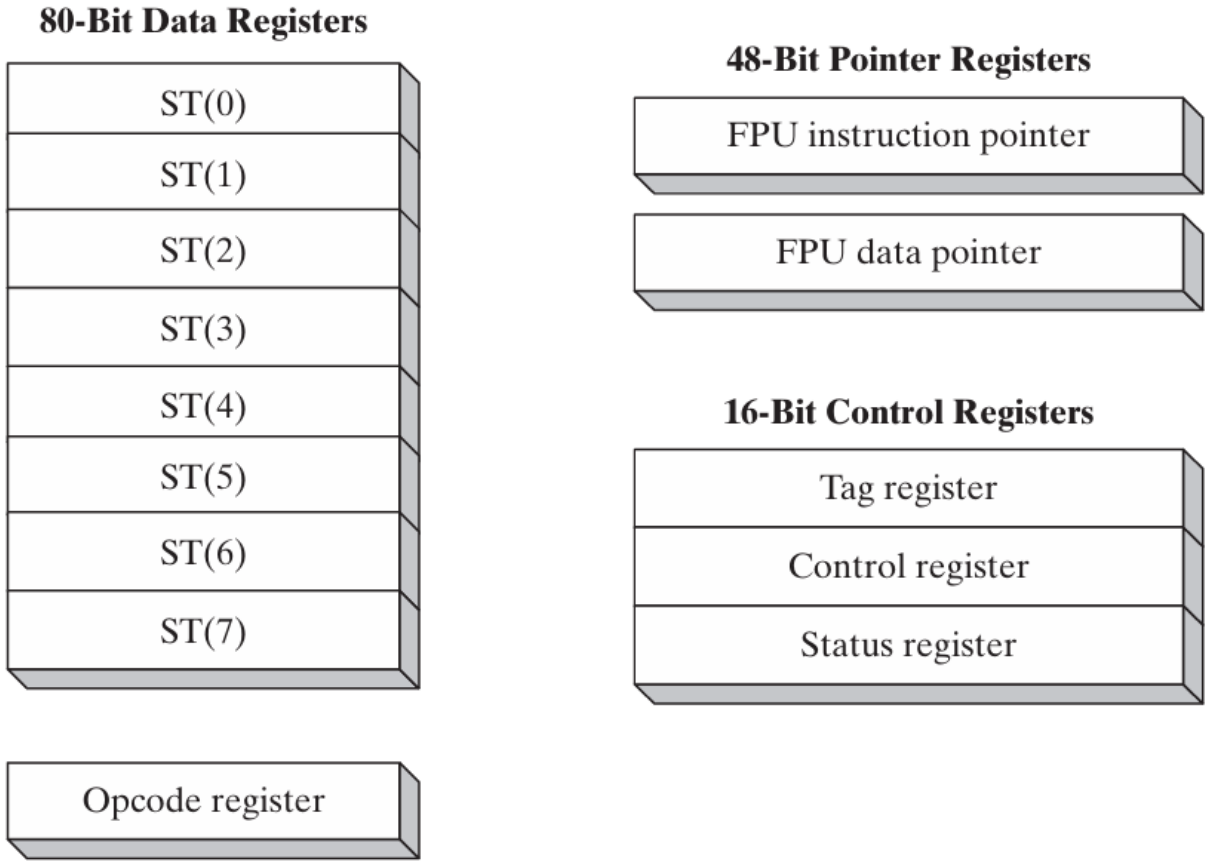
FIGURE 2–5 Floating-point unit registers.



x86 Processor Modes Comparison

Characteristic	Real-Address Mode	Protected Mode	Virtual-8086 Mode
Memory Access	Direct access to any location	Restricted access, memory protection	Simulates real mode within protected
Memory Limit	1 MB	4 GB per process	1 MB virtual space
Program Execution	Single program at a time	Multiple programs simultaneously	Multiple virtual machines
OS Support	MS-DOS, Windows 95/98	MS-Windows, Linux	Windows NT/2000/XP (for DOS)
Hardware Access	Direct access	Restricted access	Limited, some programs may not run

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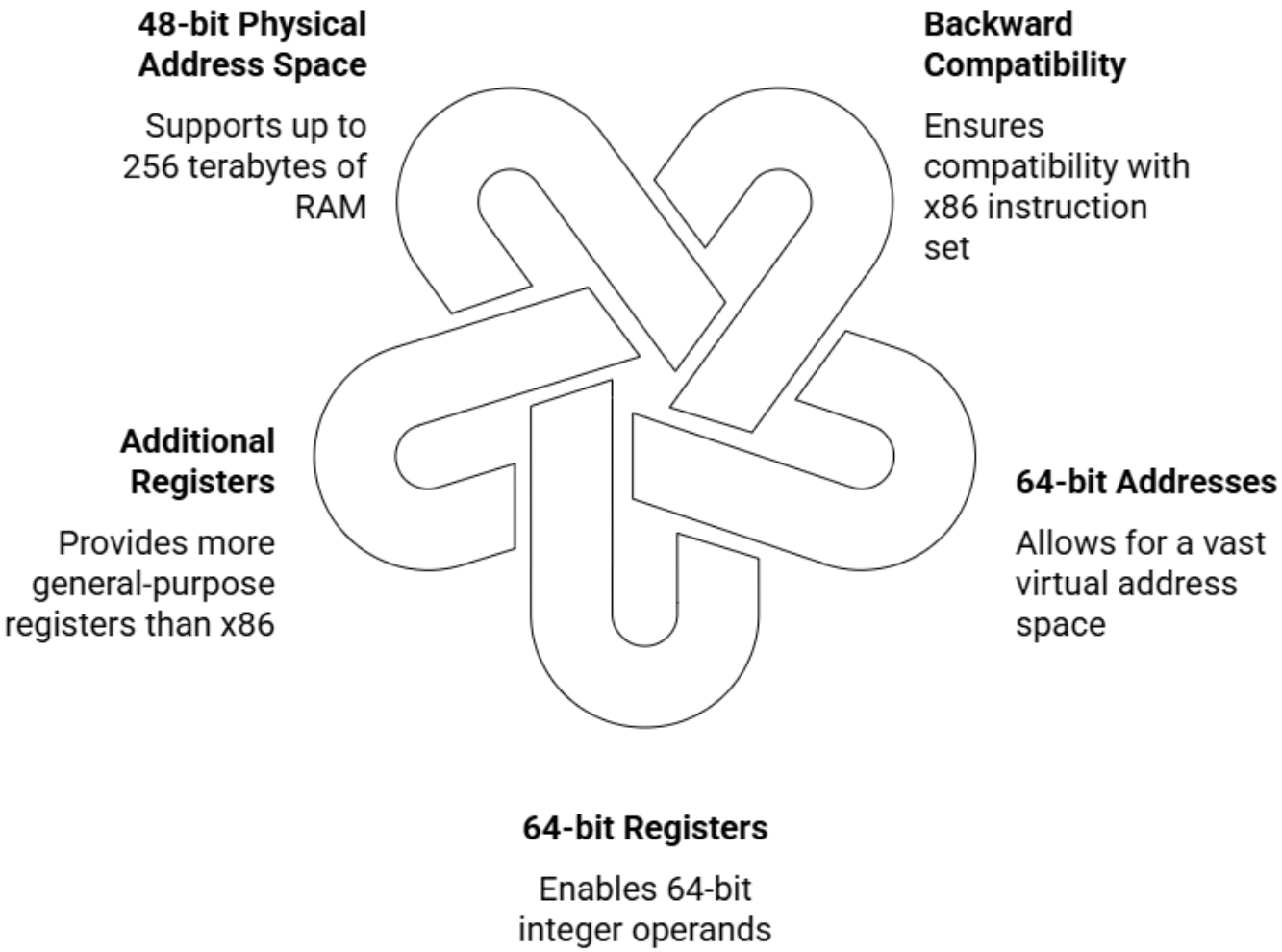
64-Bit x86-64 Processors

Key Features of x86-64 Processors



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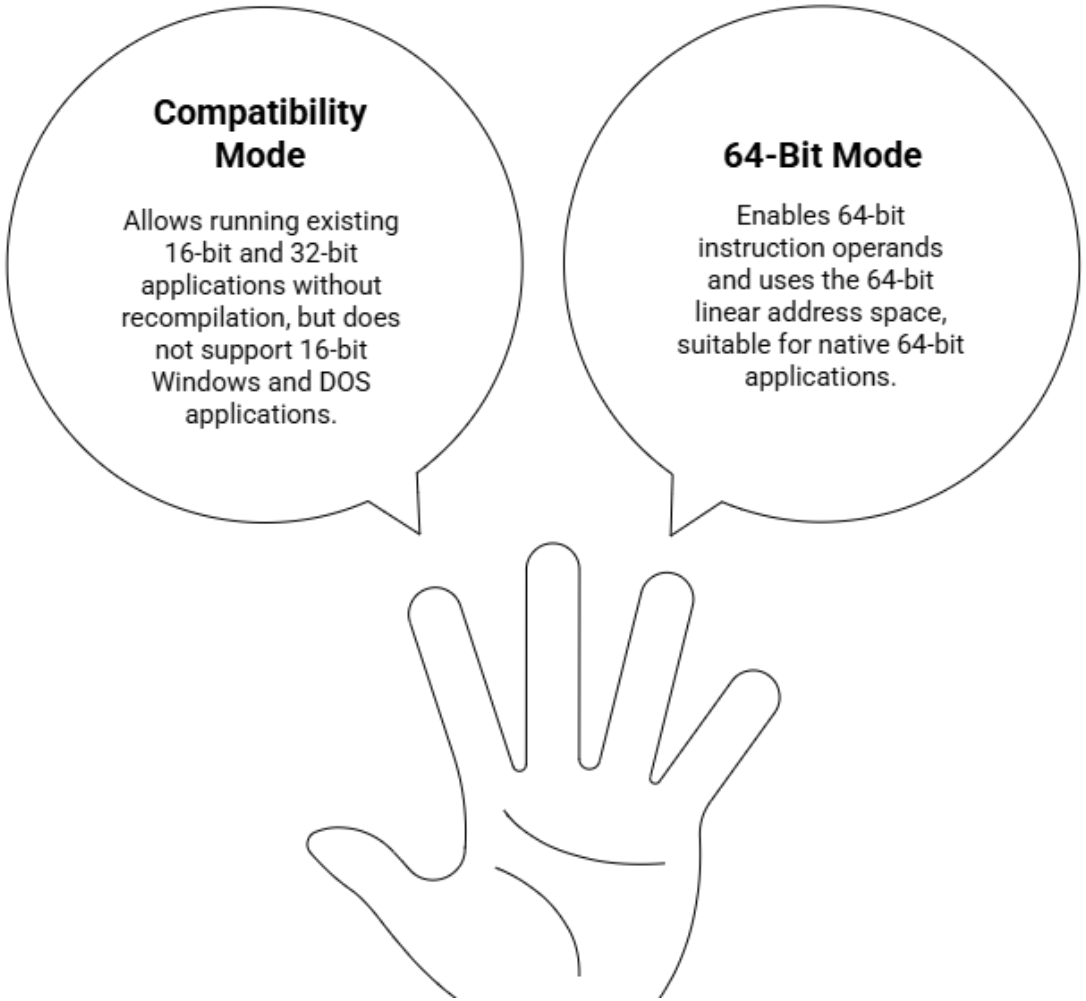
64-Bit Operation Modes - IA-32e



32-bit vs 64-bit Processor Differences

Feature	32-bit	64-bit
Address Size	32 bits	48 bits (actual) / 64 bits (theoretical)
General Purpose Registers	Eight	Sixteen
Floating-Point Registers	Eight 80-bit	Eight 80-bit
Status Flags Register	32-bit (EFLAGS)	64-bit (RFLAGS)
Instruction Pointer	32-bit (EIP)	64-bit (RIP)
MMX Registers	Eight 64-bit	Eight 64-bit
XMM Registers	Eight 128-bit	Sixteen 128-bit

General Purpose Registers

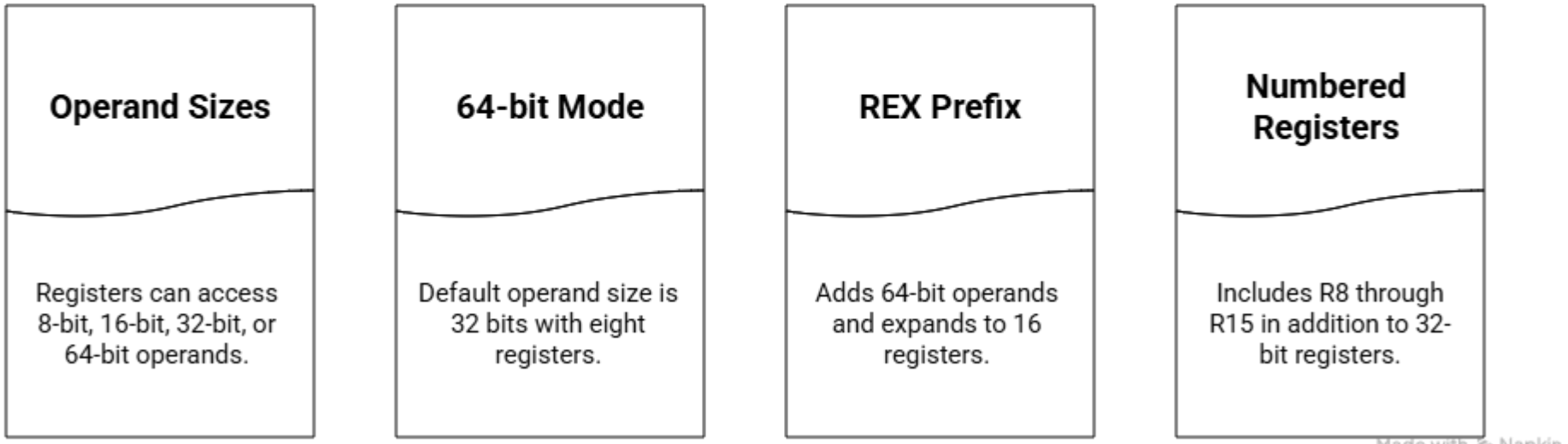


TABLE 2-1 Operand Sizes in 64-Bit Mode When REX Is Enabled.

Operand Size	Available Registers
8 bits	AL, BL, CL, DL, DIL, SIL, BPL, SPL, R8L, R9L, R10L, R11L, R12L, R13L, R14L, R15L
16 bits	AX, BX, CX, DX, DI, SI, BP, SP, R8W, R9W, R10W, R11W, R12W, R13W, R14W, R15W
32 bits	EAX, EBX, ECX, EDX, EDI, ESI, EBP, ESP, R8D, R9D, R10D, R11D, R12D, R13D, R14D, R15D
64 bits	RAX, RBX, RCX, RDX, RDI, RSI, RBP, RSP, R8, R9, R10, R11, R12, R13, R14, R15

Components of a typical x86 computer

Components of a Typical x86 Computer

Motherboard Configuration

The physical layout and components of the motherboard

Memory

The system's storage and retrieval mechanisms

I/O Ports

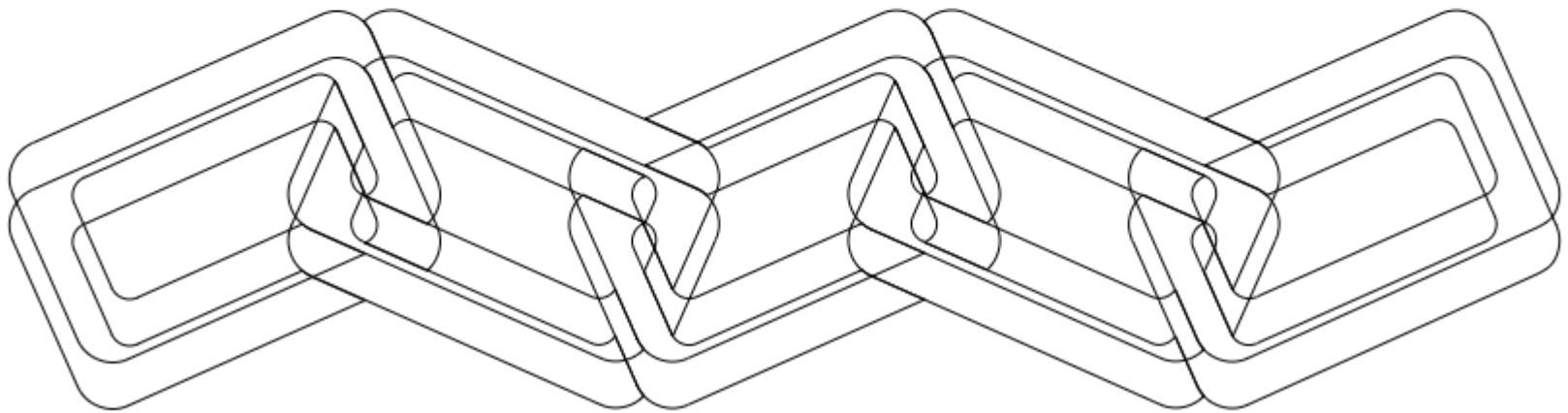
Interfaces for connecting external devices

Device Interfaces

Standards for communication between devices

Assembly Language I/O

Programming techniques for interacting with hardware



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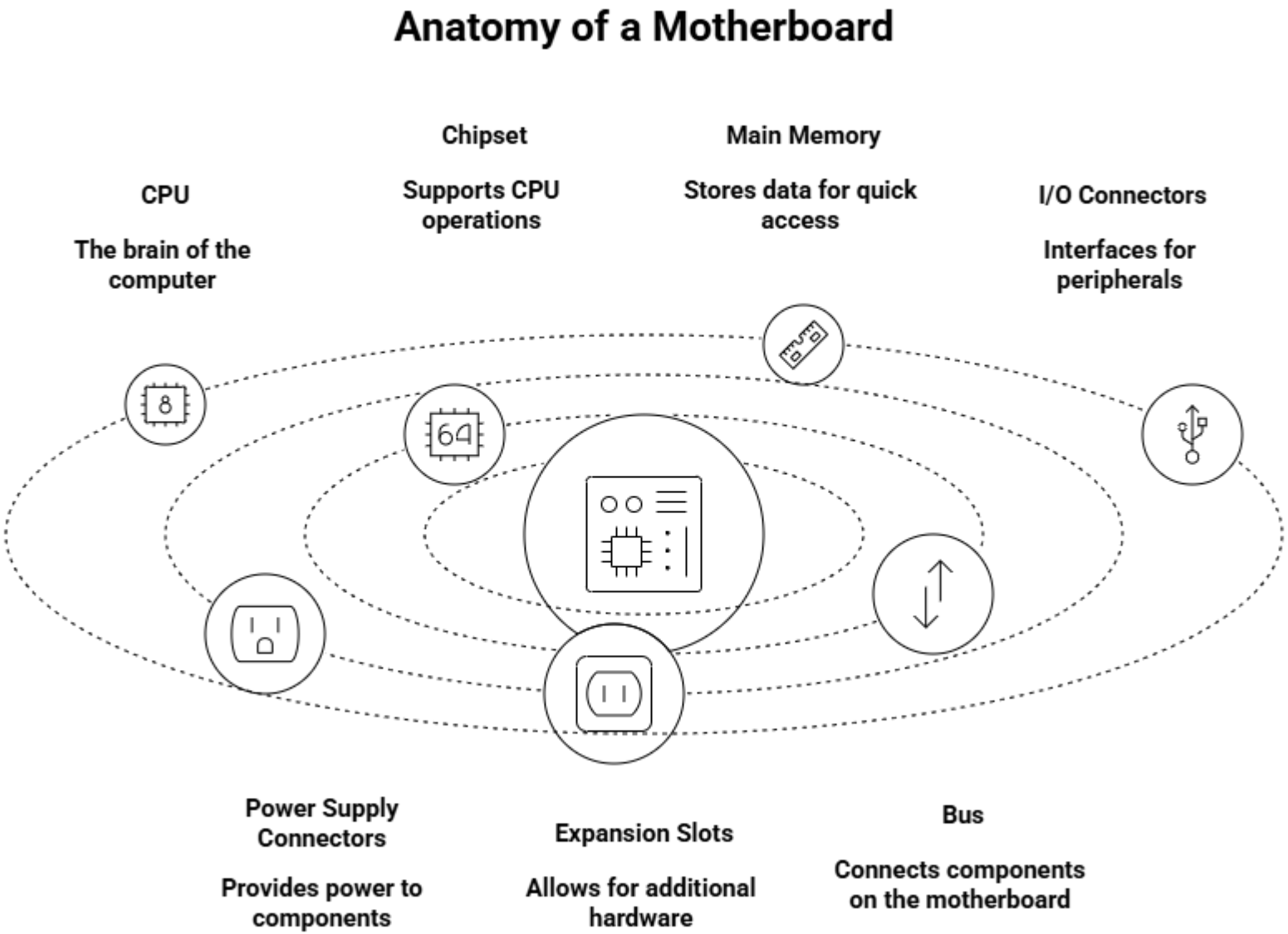
Components of a typical x86 computer



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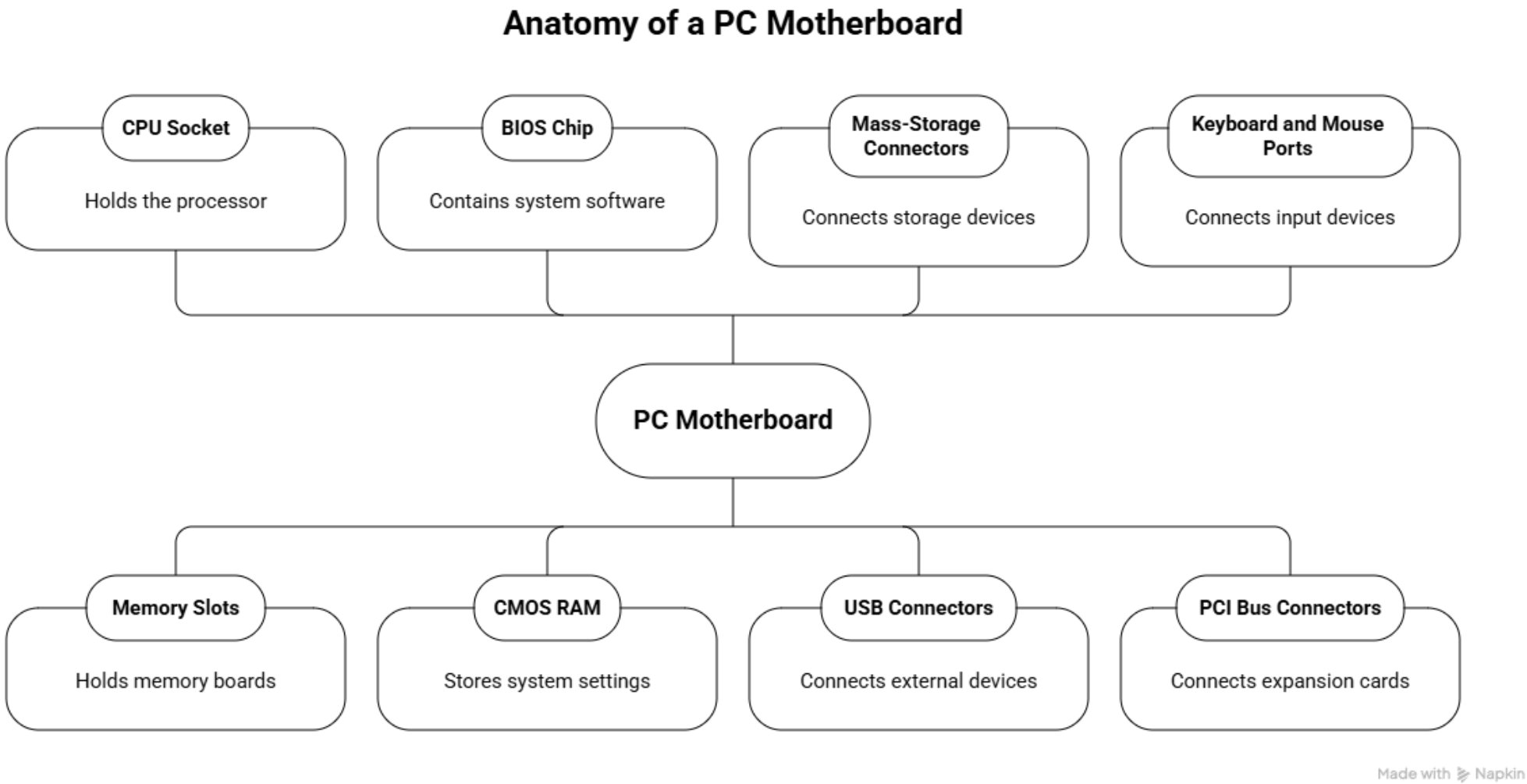
Components of a typical x86 computer



Components of a typical x86 computer

Device Connectors

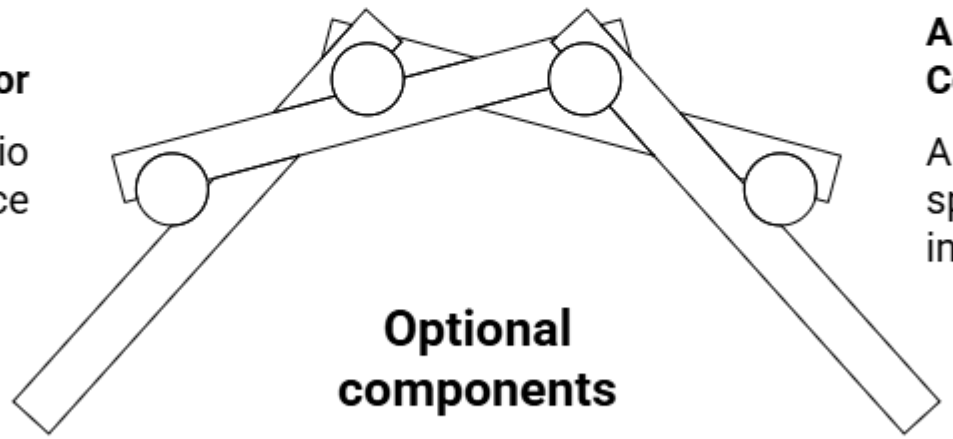
Provides versatile connectivity options

Network Adapter

Facilitates seamless internet access

Sound Processor

Enhances audio performance



AGP Bus Connector

Allows for high-speed video card installation

Optional components

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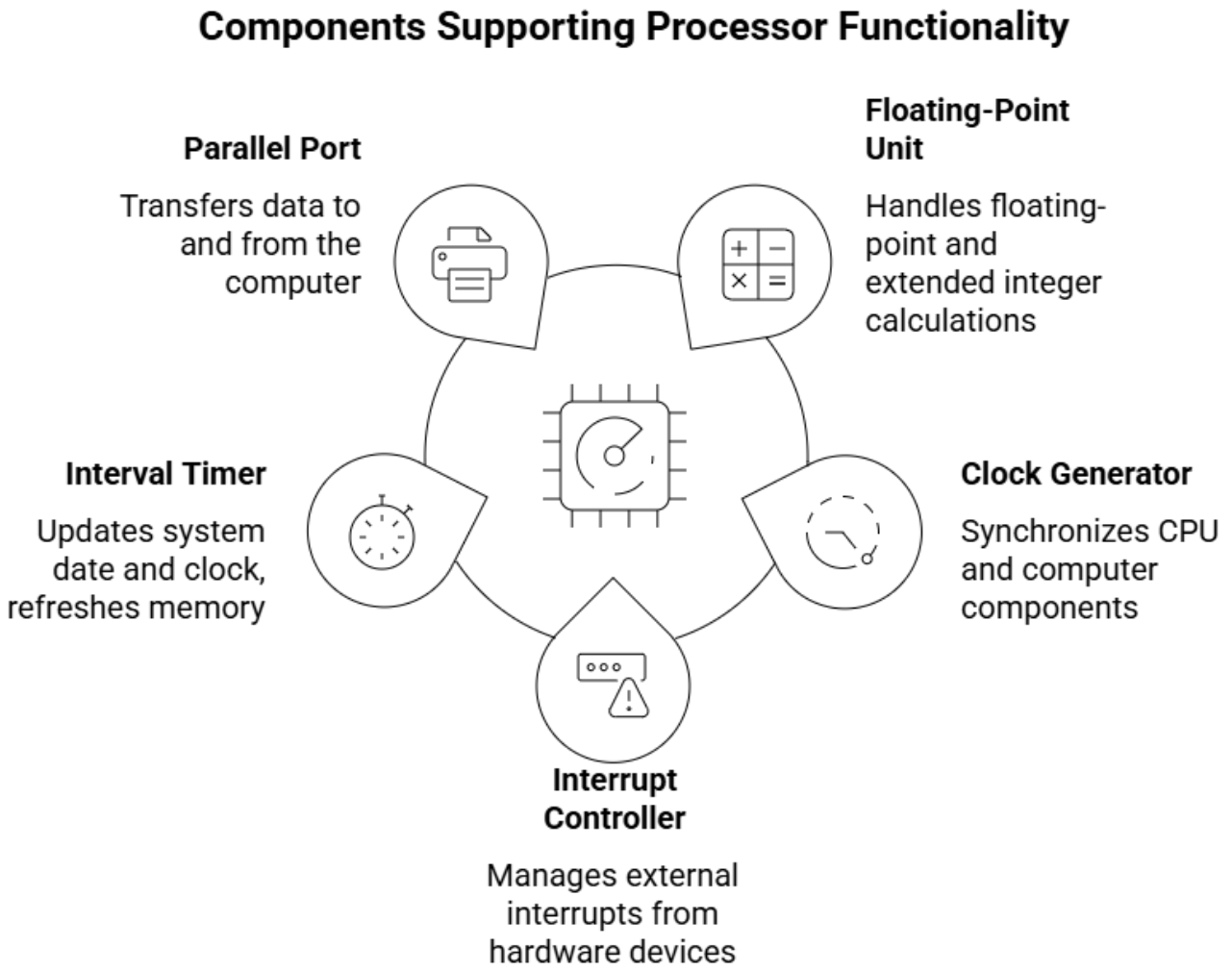
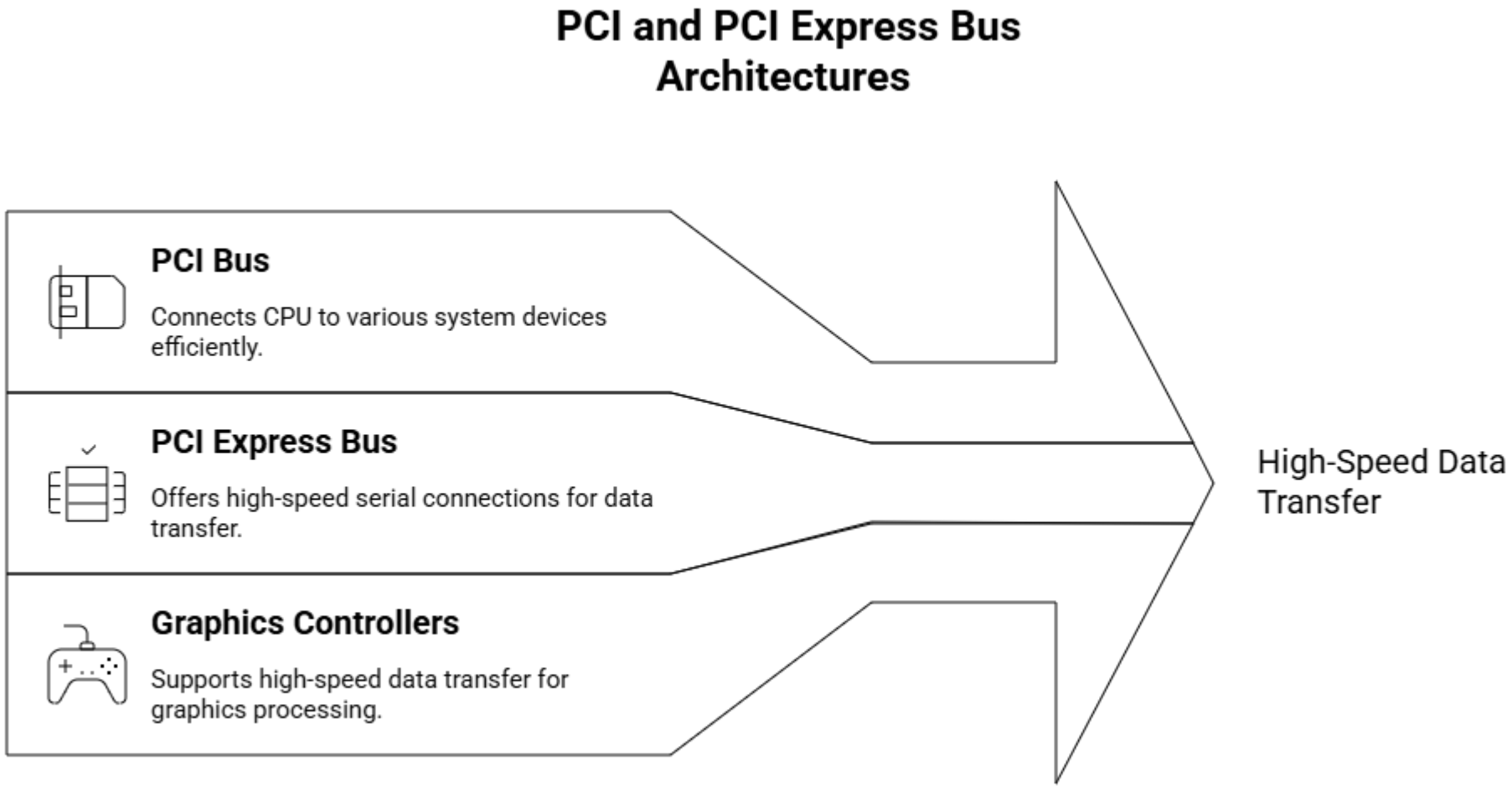
Components of a typical x86 computer



Components of a typical x86 computer



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Components of a typical x86 computer

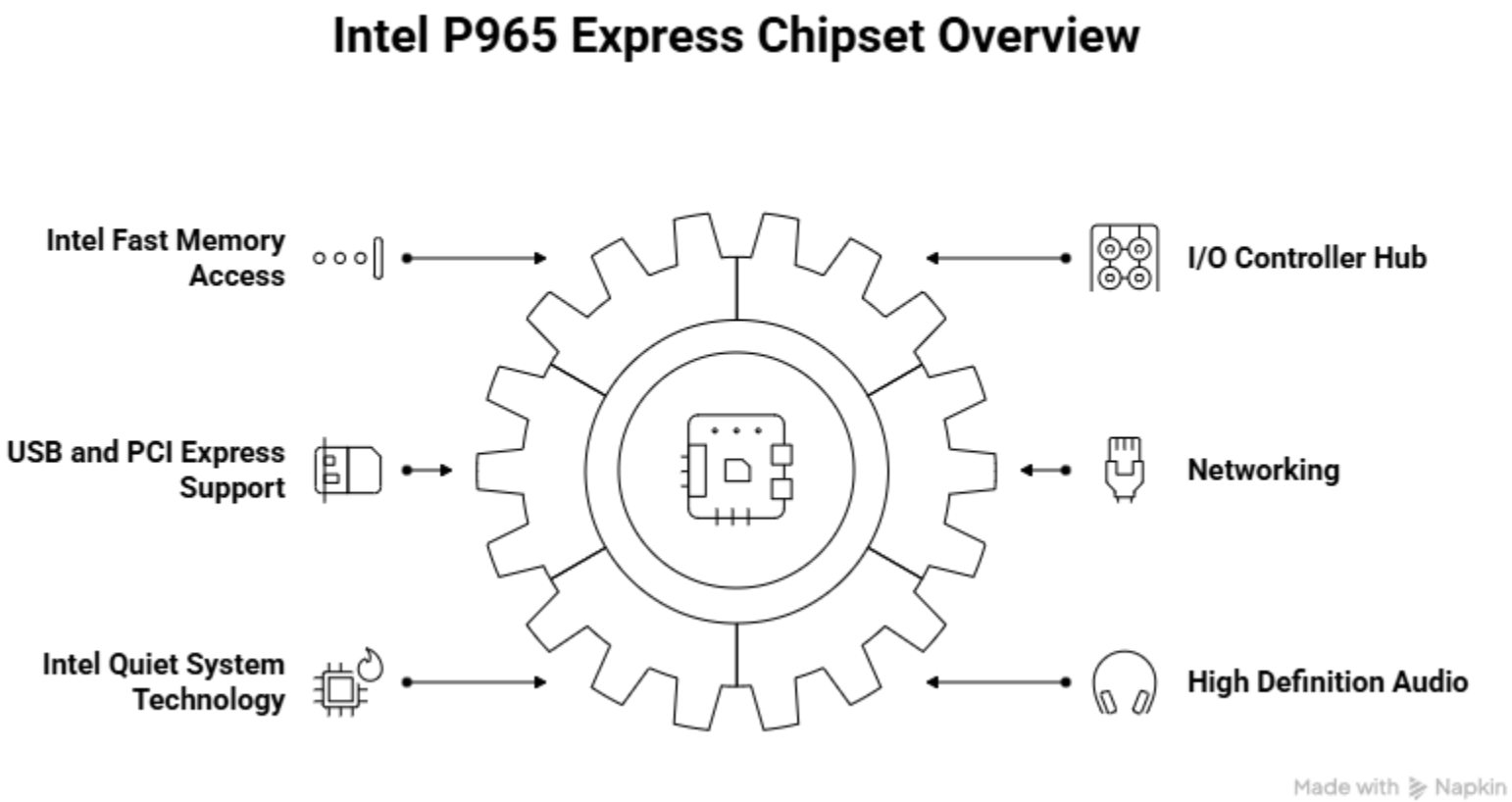
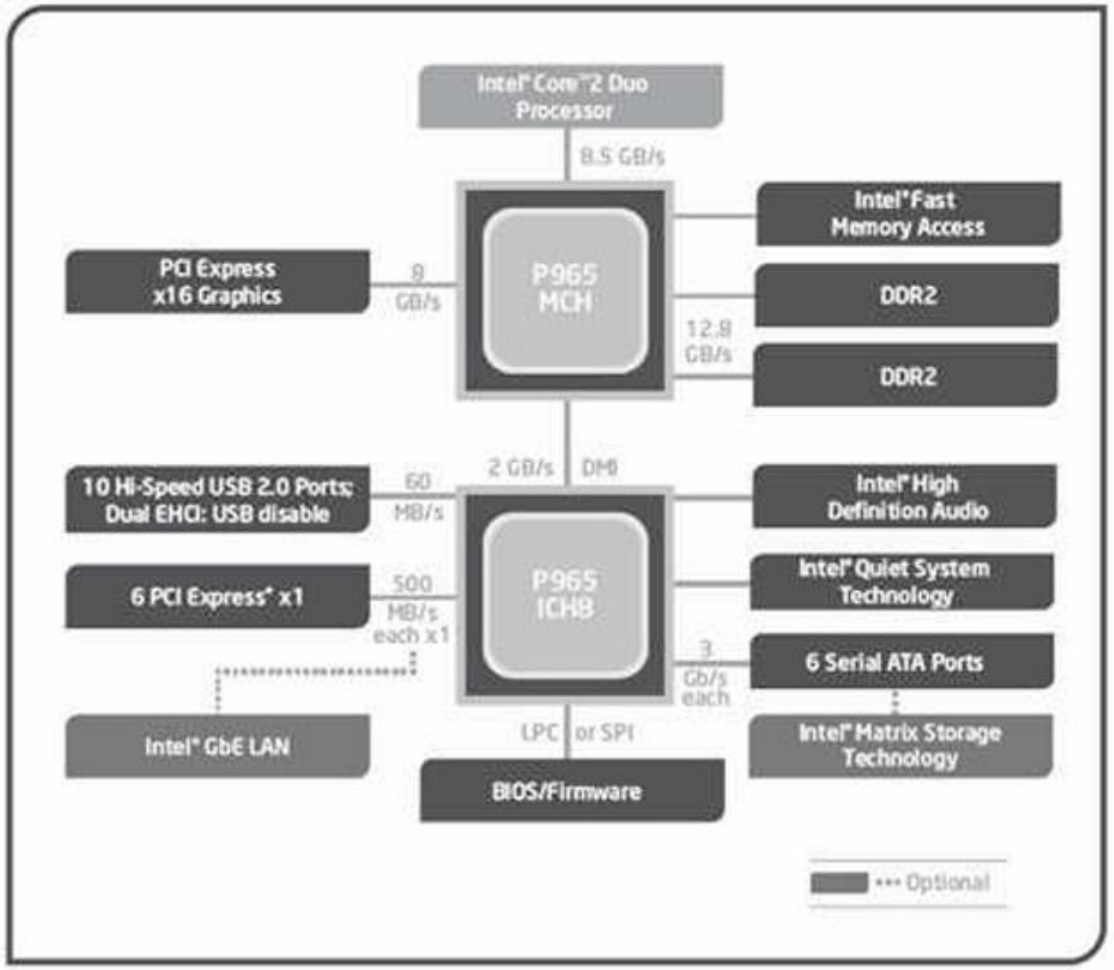
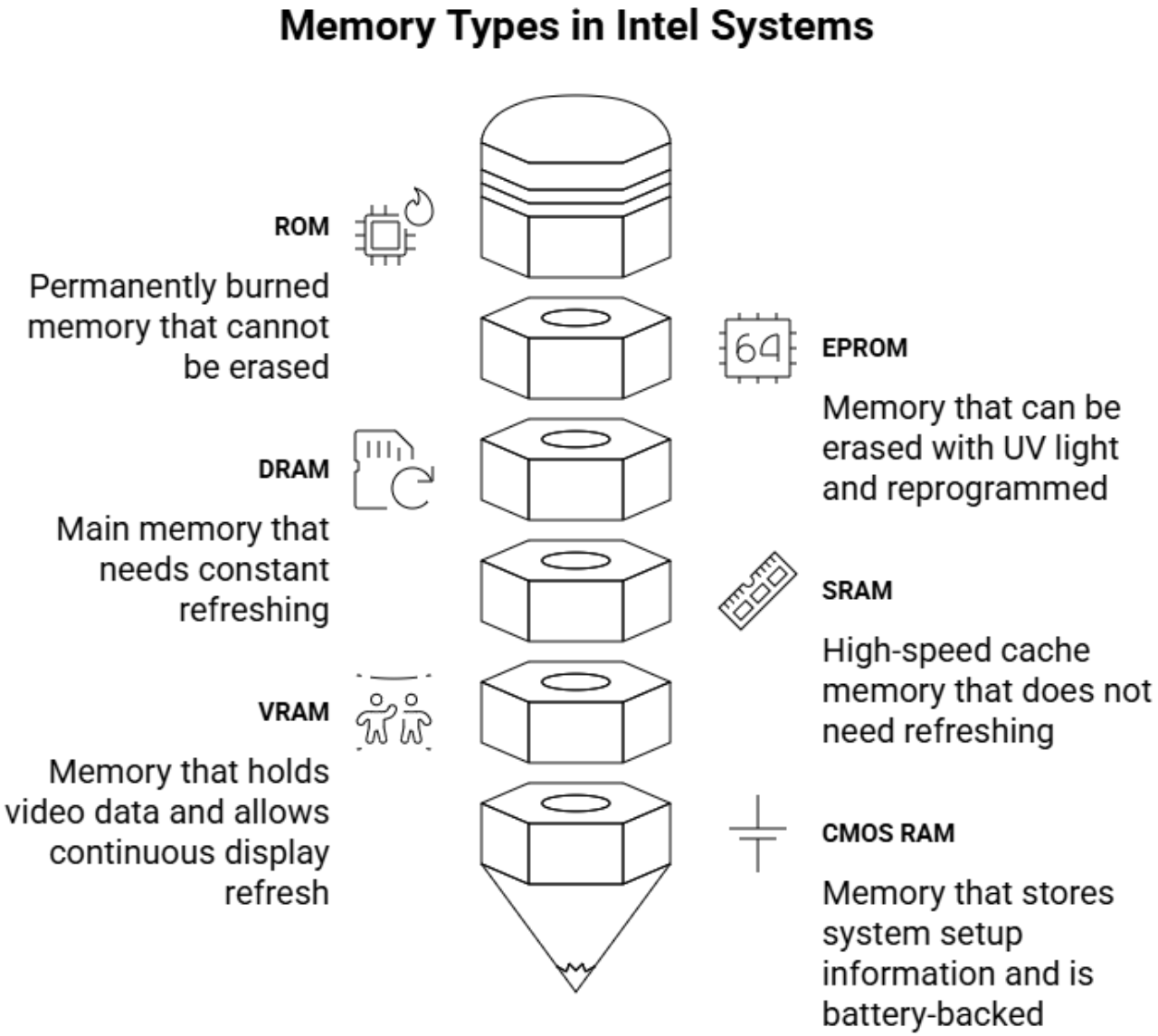


FIGURE 2–6 Intel 965 express chipset block diagram.

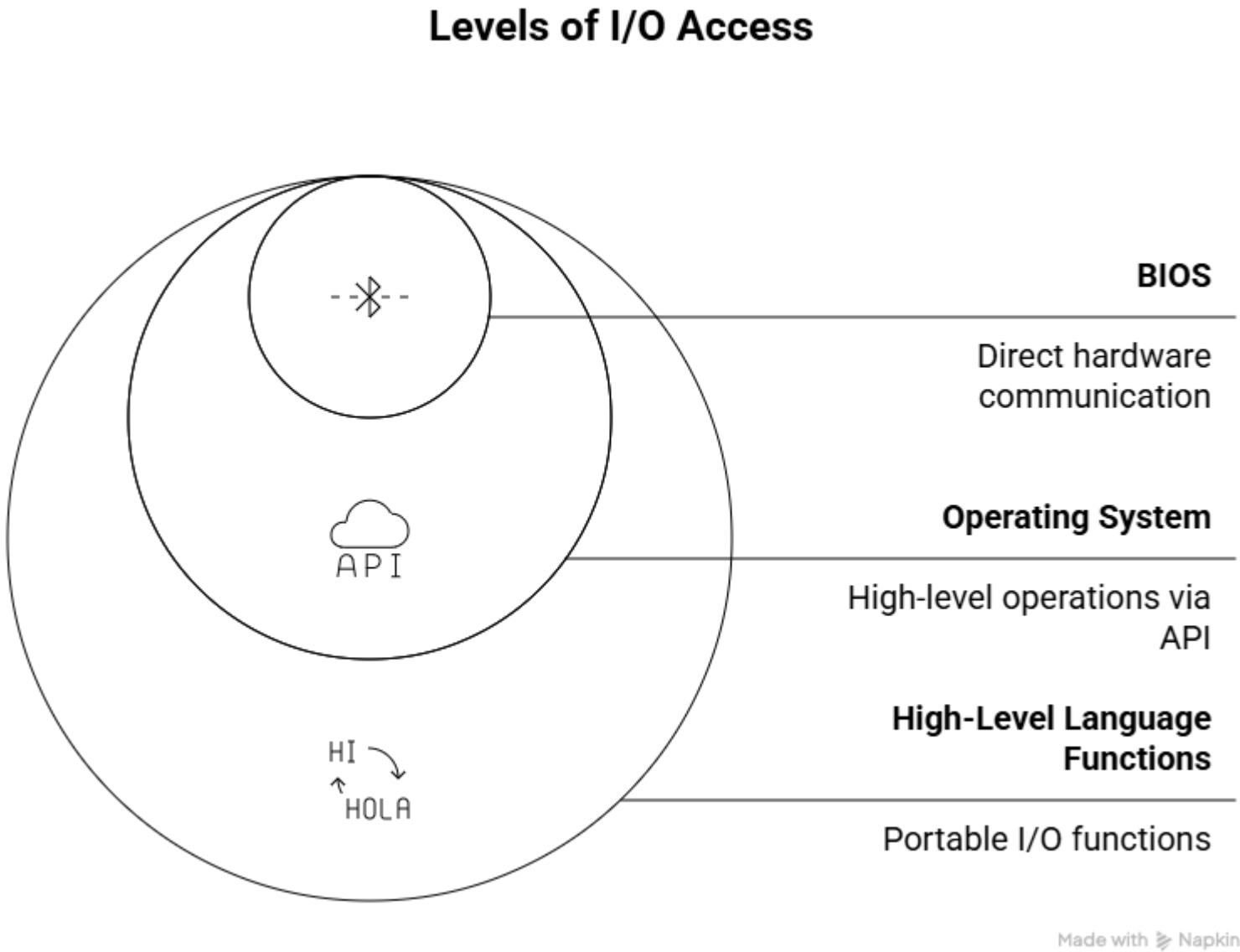


Source: The Intel P965 Express Chipset (product brief),
© 2006 by Intel Corporation, used by permission.

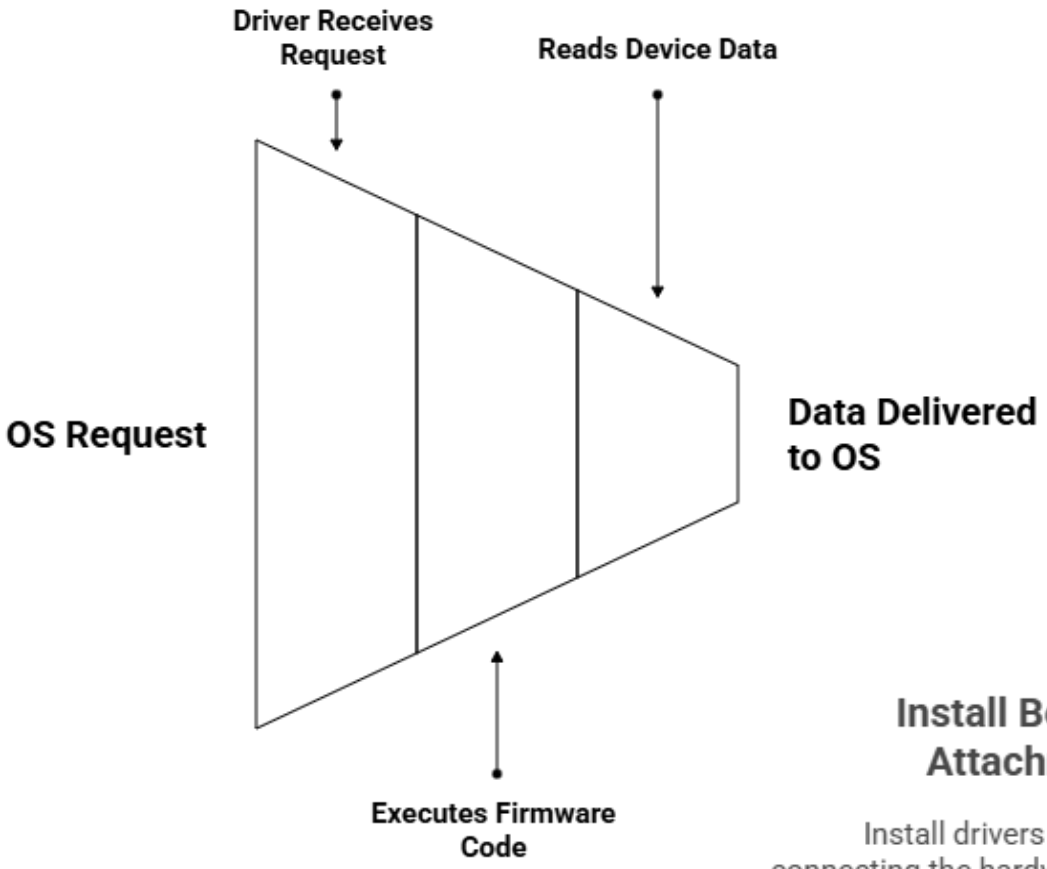
Components of a typical x86 computer



Components of a typical x86 computer



Device Driver Communication Process



How to install device drivers?

Install Before Attachment

Install drivers before connecting the hardware to ensure immediate recognition.

Install After Attachment

Install drivers after the device is connected, allowing the OS to identify and install automatically.

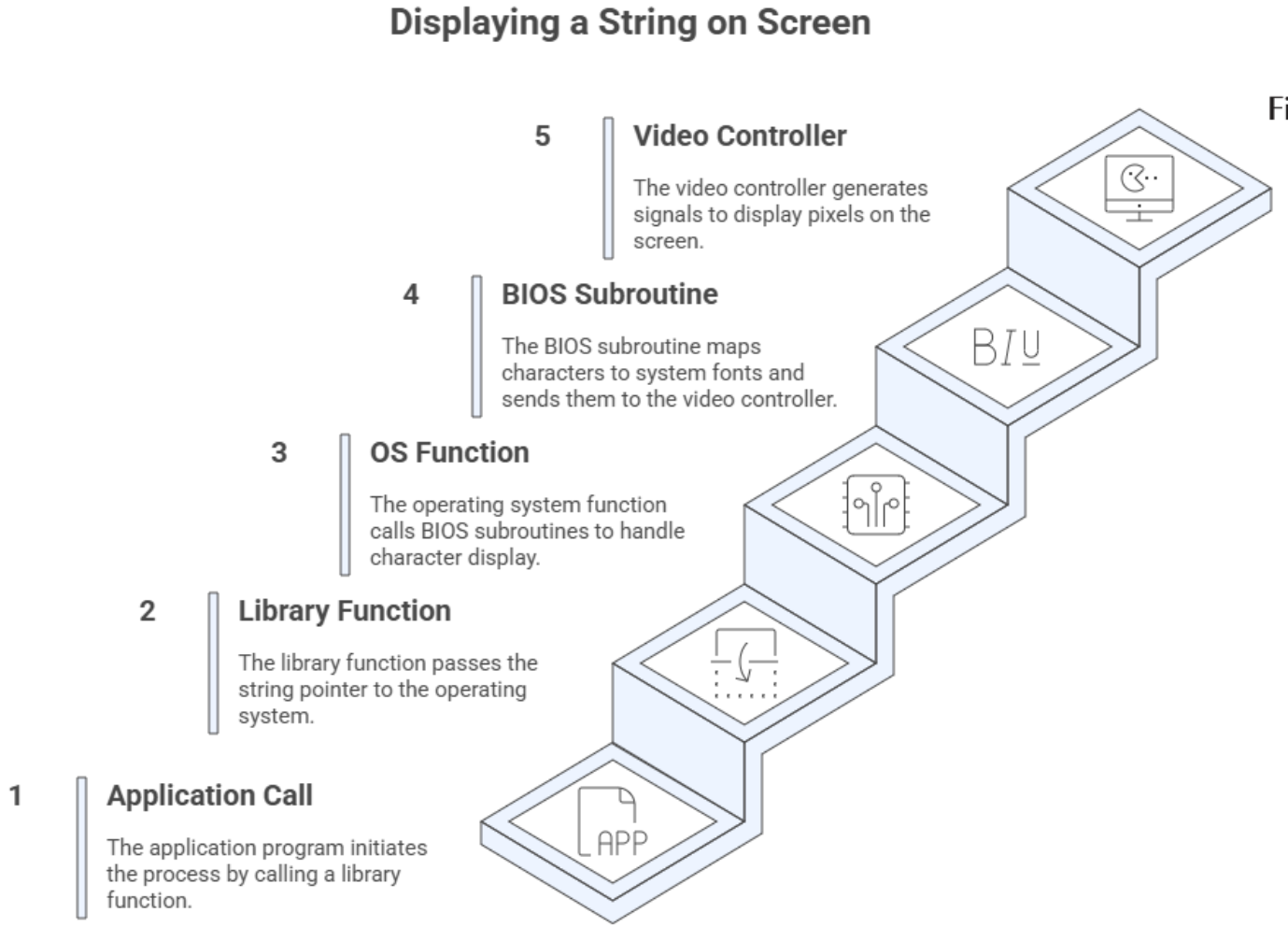
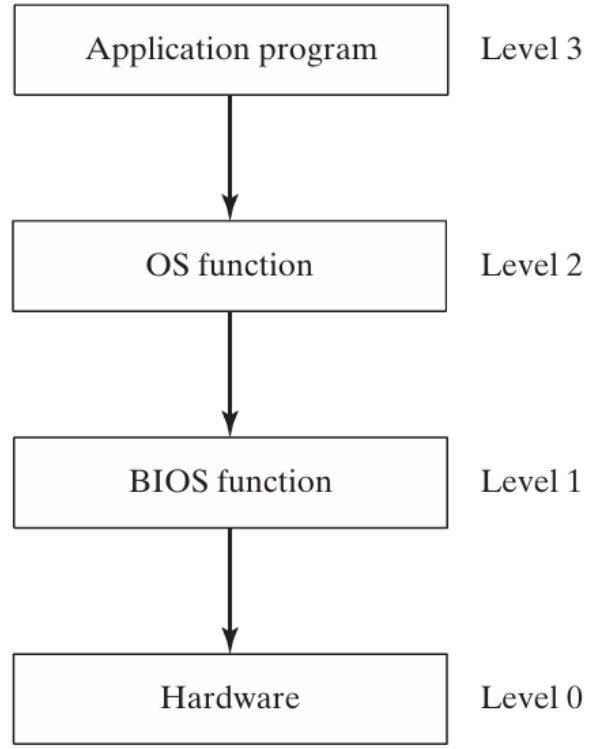


Figure 2–7 Access levels for input–output operations.



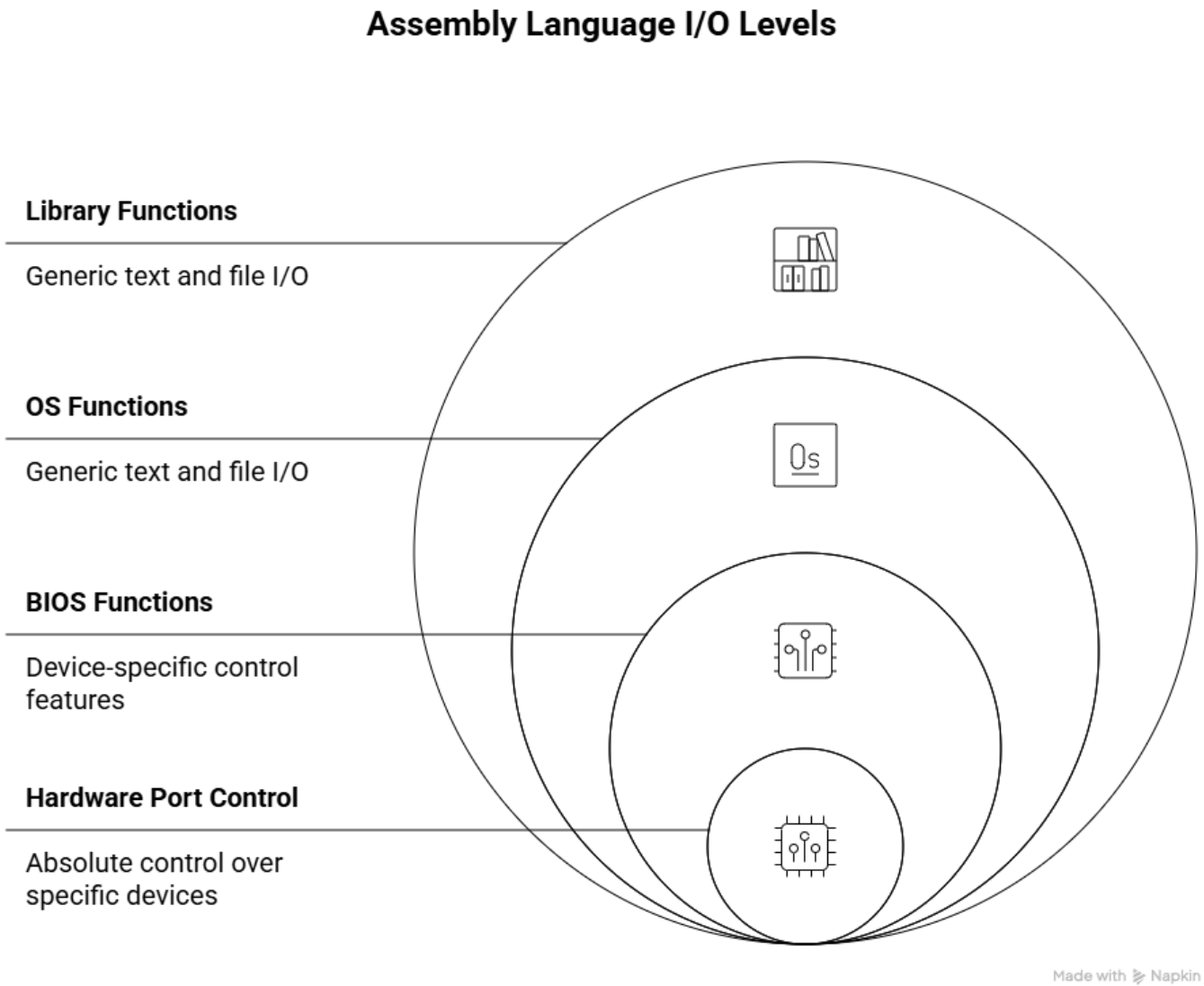
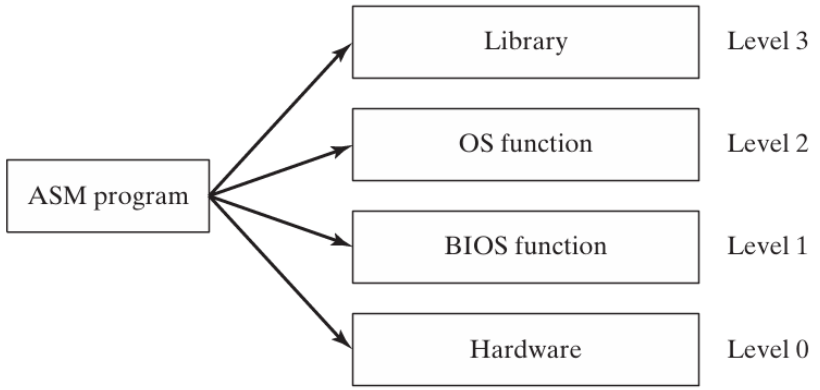


FIGURE 2–8 Assembly language access levels.



Input-output system

chapter summary

Review questions

Github upload

Presentation subject

- Motherboard
- CPU & GPU
- Main memory & SSD
- Keyboard, Mouse, Monitor
- PC building